

Words Speak Louder without Actions

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ABSTRACT

Information and control rights are central aspects of leadership, management, and corporate governance. This paper studies a principal-agent model that features both communication and intervention as alternative means to exert influence. The main result shows that a principal's power to intervene in an agent's decision limits the ability of the principal to effectively communicate her private information. The perverse effect of intervention on communication can harm the principal, especially when the cost of intervention is low or the underlying agency problem is severe. These novel results are applied to managerial leadership, corporate boards, private equity, and shareholder activism.

Actions speak louder than words, but not nearly as often.

—Mark Twain

INFORMATION AND CONTROL RIGHTS ARE central aspects of leadership, management, and corporate governance. In practice, communication of private information and intervention in the decision-making process are common remedies for information asymmetries and conflicts of interest in a wide range of situations. The interplay between communication and intervention, however, is little understood. In this paper, I show that the power of a principal to intervene in an agent's decision exacerbates the underlying agency problem and as a result limits the ability of the principal to use her private information to influence the agent's decision. The power to intervene can therefore be detrimental to the principal. This novel result has implications for the effectiveness of visionary management, the tension between the supervisory and advisory roles of corporate boards, and the value that sophisticated investors offer their portfolio companies.

To study the interaction between communication and intervention, I consider a principal-agent model with incomplete contracts and a “top-down”

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information structure. In the model, the optimal scale of investment depends on the fundamentals of the firm. The principal, who is privately informed about these fundamentals, sends the agent a message that can be interpreted as a nonbinding demand. Communication is modeled as cheap talk as in Crawford and Sobel (1982). The agent has a tendency to overinvest (e.g., empire-building) and thus the challenge facing the principal is convincing the agent to choose small projects when firm fundamentals are bad. In equilibrium, information is never fully revealed by the principal, who has incentives to understate firm fundamentals to prevent overinvestment. The novel feature of the model is the possibility of intervention. Specifically, after communicating with the agent, the principal observes the agent's decision and decides whether to intervene and adjust the size of the project. For example, the principal can overrule the agent or monitor him more closely. Since these activities require resources and attention, intervention is costly to the principal.

The main result of the paper is that intervention hinders communication. In equilibrium, less information may be revealed by the principal if she has the power to intervene in the agent's decision. The power to intervene therefore limits the ability of the principal to influence the agent. Since communication is more effective without intervention, *words speak louder without actions*.

How can intervention hinder communication? The underlying mechanism rests on the limited commitment of the principal and has two related channels. First, in equilibrium, the principal intervenes to alleviate the overinvestment problem. However, since intervention is costly, it is never in the principal's best interest to completely undo the agent's bias. In general, more overinvestment warrants more intervention, but it ultimately results in a larger final project. The agent anticipates the principal's intervention. In response, he deliberately chooses projects that are larger than what he would have preferred in the absence of intervention. By overshooting, the agent increases the cost the principal must incur to downsize the project, thereby guaranteeing the desired amount of overinvestment. Effectively, the agent behaves as if his bias toward overinvestment is larger, and as a result, the principal has even stronger incentives to understate the true fundamentals of the firm. In other words, the principal's attempt to prevent the agent from undoing her expected intervention further diminishes her credibility when communicating with the agent. This "vicious cycle" contributes to less informative communication in equilibrium.

Second, intervention hinders communication to the extent that it is also an alternative channel through which the agent can "elicit" private information from the principal. To understand this channel, note that because of the intrinsic conflict of interest, the principal never fully communicates her information in equilibrium. Therefore, the agent always faces uncertainty about the fundamentals of the firm. In general, the risk of making a large investment when fundamentals are bad reduces the agent's incentives to overinvest. Intervention weakens this mitigating force by providing the agent with an alternative source of information. Indeed, the agent can condition his decision on the information embedded in the principal's decision to intervene, information that

was not previously communicated. Knowing that the principal will intervene only when fundamentals are very bad emboldens the agent to take more “risk” by choosing larger projects than he would take otherwise. Put differently, to elicit additional information from the principal, the agent must provoke intervention, which he does by intentionally overshooting. Since the principal expects the agent to be less responsive to her messages, she has stronger incentives to understate the true fundamentals and hence communication is less informative in equilibrium.

The perverse effect of intervention on communication is particularly strong when the cost of intervention is low or the underlying agency problem is severe. Intuitively, the principal intervenes more aggressively when the cost is lower. Therefore, more overshooting by the agent is needed to suppress the impact of intervention on the final project. In addition, since more aggressive intervention more fully reveals the principal’s private information, the risk of overinvesting when fundamentals are bad is lower, which induces the agent to choose even larger projects in equilibrium. The intrinsic bias of the agent toward overinvestment has a similar effect. A larger bias implies more overinvestment, and since intervention is more beneficial to the principal when overinvestment is detrimental, a larger bias prompts the principal to intervene more aggressively. Similar to the reasoning behind the effect of the cost of intervention, the agent has even stronger incentives to overshoot, further impeding effective communication.

The perverse effect of intervention on communication is detrimental—it can offset the value of intervention as a correction device and reduce the principal’s expected welfare, especially when the cost of intervention is low or the underlying agency problem is severe. In other words, from the perspective of the principal, the power to intervene is least desirable when intervention is seemingly most effective or most needed. As explained above, under these circumstances, the negative effect of intervention is particularly strong, and as a result it dominates the other benefits of intervention. The idea that communication in and of itself can reduce the value of control rights is another novel aspect of the analysis.¹ Interestingly, when intervention harms the principal, it benefits the agent. Similar to the principal, the agent suffers from the negative effect of intervention on communication. However, as intervention also provides an informational benefit to the agent, it can substitute for the lack of informative communication and offset its negative effect on the agent.

I consider several extensions to the baseline model. First, I demonstrate that intervention hinders communication and decreases the principal’s welfare even if the principal can choose the sensitivity of the agent’s payoff to performance. Second, intervention has a weaker adverse effect on communication if

¹ This result does not imply that the principal is worse-off with communication. As in Crawford and Sobel (1982), the sender (i.e., the principal) is ex ante better-off when more information is communicated in equilibrium. This is true whether or not the sender can intervene. See Melamud and Shibano (1991) for an alternative cheap-talk game in which the sender can be ex ante better-off in equilibrium without communication.

it imposes a direct cost on the agent (e.g., loss of compensation, damaged reputation, or embarrassment). Intuitively, a direct cost of intervention reduces the incentives of the agent to overshoot. As a result, the principal has fewer incentives to understate the fundamentals of the firm, and more information can be revealed in equilibrium. If this cost is sufficiently large, then the agent may even undershoot, in which case intervention facilitates communication. Finally, I demonstrate that the adverse effect of intervention on communication also holds when both the principal and the agent are privately informed about the fundamentals of the firm.

Building on these insights, I discuss novel implications of the analysis for managerial leadership, corporate boards, private equity (PE), and shareholder activism. In all of these applications, communication and intervention are the primary governance mechanisms. For example, the model predicts that visionary leadership and a “hands-off” managerial style are more likely to be successful in large and complex organizations. In addition, the advisory role of corporate boards is expected to be more significant when the number of directors is large or when directors are diverse and busy (e.g., hold other board seats). The model also suggests that PE investors’ voices are more likely to be heard when their investment is cosponsored or when they have better exit options (e.g., booming IPO and M&A markets). Similarly, the ease with which activist hedge funds can launch a proxy fight could actually decrease their ability to influence the policy of their target companies.

This paper is related to the literature on incomplete contracts. Aghion and Tirole (1997) study a model with intervention and communication, and show that the power to overrule the agent is detrimental because it weakens the agent’s incentives to acquire information.² This hold-up problem is absent from my model. Aghion and Tirole (1997) also assume that the uninformed agent always follows the recommendations of the principal,³ and as a result their model predicts that the allocation of control is irrelevant when the agent is uninformed. In contrast, my model predicts that the principal’s commitment not to intervene can be strictly optimal. In this respect, my model shares with Crémer (1995) the idea that the principal can benefit from letting the agent live with the consequences of his actions. However, the mechanisms here are quite different. First, in Crémer’s (1995) model, the principal benefits from being uninformed (about the agent’s ability) because this results in more “aggressive” firing decisions that incentivize the agent to exert more effort.⁴ By contrast, in my model letting the agent live with the consequences of his actions literally means that the principal (who is always informed) commits not to intervene

² Similarly, Burkart, Gromb, and Panunzi (1997) show that intervention undermines managerial initiatives, and Adams and Ferreira (2007) show that it disincentivizes CEOs from cooperating with their board.

³ In their model, the uninformed agent prefers the principal’s most favored project over the risk of choosing a project with a “sufficiently negative” payoff, which by assumption always exists.

⁴ The firing decisions of the uninformed principal in Crémer (1995) are more aggressive in the sense that the agent is fired if and only if output is low, irrespective of his true ability.

in the agent's decision.⁵ Second, since in Crémer (1995) the principal is initially uninformed, communication plays no role in his analysis. Importantly, this strand of the literature is silent about the effect of intervention on the quality of communication, which is the main focus of my analysis. Studying the interplay between these two mechanisms is thus empirically relevant, and highlights a novel mechanism through which the allocation of control rights affects real outcomes.

This paper is also related to the literature on delegation. Starting with Dessein (2002), a number of papers study the trade-off between delegation and communication in organizations,⁶ and in particular, its applications to optimal board structure (Adams and Ferreira (2007), Harris and Raviv (2008), Chakraborty and Yilmaz (2017)) and shareholder control (Harris and Raviv (2010)). In these models, the uninformed principal delegates decision rights to the informed agent, and delegation is beneficial because it avoids the distortion of the agent's private information when he communicates with his principal. By contrast, in my model the informed principal communicates with the uninformed agent. The trade-off is between backing this communication with intervention and relying solely on communication as a governance mechanism. Moreover, while the papers above imply that the benefit to the principal from retaining decision-making authority is greater when communication (by the agent) is allowed, my analysis suggests that communication by the principal in and of itself can reduce the value of control rights. In this respect, my model offers new implications for corporate governance.

Related, Matthews (1989) studies a model in which the principal has the right to veto the agent's decision following cheap-talk communication. Importantly, in his model the principal's private information is about her preferences, not the common value of the project/task. Therefore, the agent values this information only to the extent that it affects the principal's decision to exercise her veto right, and as a result a veto threat can only improve communication. Shimizu (2017) analyzes a model in which the principal can exit the relationship following a cheap-talk communication with the agent. Since in Shimizu's (2017) model, exit punishes the agent but does not change his decision, exit can only improve communication. The punishment element of intervention also exists in Marino, Matsusaka, and Zábojník (2010), Van den Steen (2010), and Levit (2019), who similarly find that intervention facilitates communication.

Finally, existing models of leadership focus on the leader's role in coordinating various activities of the firm (e.g., Hermalin (1998), Bolton, Brunnermeier, and Veldkamp (2013)).⁷ This paper contributes to this literature by showing that the ease with which corporate leaders can exercise their power decreases their ability to influence others to follow their vision.

⁵ In Crémer (1995), the principal does not benefit from forgoing the right to fire the agent.

⁶ For example, see Agastya, Bag, and Chakraborty (2014), Alonso and Matouschek (2007), Grenadier, Malenko, and Malenko (2016), Mylovanov (2008), and Harris and Raviv (2005).

⁷ Rotemberg and Saloner (1993, 2000) also focus on the vision aspect of leadership, but without modeling top-down communication. See Bolton, Brunnermeier, and Veldkamp (2010) for a related survey.

The reminder of the paper is organized as follows. Section I introduces the baseline model, characterizes the equilibrium, presents the main result, and describes the welfare implications of the analysis. Section II considers several extensions to the baseline model. Section III discusses the novel implications of the analysis. Section IV concludes. All proofs and supplemental results are provided in the Appendix and the Internet Appendix, respectively.⁸

I. Analysis

This section presents the paper's main analysis. In Section I.A, I begin by introducing the model, and in Section I.B, I characterize the equilibrium. In Section I.C, I study the effect of intervention on communication, and in Section I.D, I consider welfare implications.

A. Model Setup

I start by describing the primitives of the model, which are standard in the literature. The possibility of intervention, which is the key innovation of the model, is introduced toward the end of this subsection.

Consider a principal-agent environment in which payoffs depend on action $x \in \mathbb{R}$ and a random variable θ that is uniformly distributed over $[\underline{\theta}, \bar{\theta}]$.⁹ The principal's payoff is given by

$$U_P(x, \theta) = U_P(\theta, \theta) - L(|x - \theta|), \quad (1)$$

where $L''(\cdot) > 0$ and $L(0) = L'(0) = 0$. Notice that $U_P(x, \theta)$ obtains its unique maximum at $x = \theta$. I refer to x as the scale of investment (or type of project) and to θ as the fundamentals of the firm (e.g., productivity). Similarly, the agent's payoff is

$$U_A(x, \theta; b) = U_A(\theta + b, \theta; b) - T(|x - \theta - b|), \quad (2)$$

where $T''(\cdot) > 0$ and $T(0) = T'(0) = 0$. Notice that $U_A(x, \theta; b)$ obtains its unique maximum at $x = \theta + b$. Without loss of generality, I assume that $b > 0$, that is, relative to the principal, the agent has incentives to overinvest. In effect, b captures the intrinsic conflict of interest between the principal and the agent, where a larger b implies a larger conflict. The conflict of interest could stem from the agent's reputation concerns, private benefits (e.g., perks, prestige, power), cost of exerting effort, or desire for the "quiet life." Following Grossman and Hart (1986) and Hart and Moore (1990), I assume that contracts are incomplete. In particular, the agent and the principal cannot contract over projects or

⁸ The Internet Appendix is available in the online version of this article on the *Journal of Finance* website.

⁹ A uniform distribution is often assumed in cheap-talk models to gain tractability (e.g., Dessein (2002), Harris and Raviv (2005, 2008, 2010), Adams and Ferreira (2007), Chakraborty and Yilmaz (2017)).

the communication protocol. In Section II.A, I discuss the effect of pay-for-performance contracts.

The sequence of events in the model is as follows. The initial stage involves communication between the principal and the agent. The principal has private information about θ . For simplicity, I assume that the principal perfectly observes θ , while the agent is uninformed. These assumptions are relaxed in Section II.C. Based on her private information, the principal sends the agent a message $m \in [\underline{\theta}, \bar{\theta}]$. The agent observes the message from the principal and chooses project $x \in \mathbb{R}$. In line with a standard cheap-talk framework, the principal's information about θ is nonverifiable, and the content of m does not directly affect the agent's or the principal's payoff. These assumptions leave room for information manipulation.

The second stage is the key departure of the model from the existing literature. The principal observes the agent's decision and then decides on $\Delta \in \mathbb{R}$, the extent of intervention. If $\Delta = 0$, the principal does not intervene and the agent's choice is implemented as is. If $\Delta \neq 0$, the principal intervenes and the final project is $x - \Delta$. The principal incurs a cost of $\delta C(|\Delta|)$, where $C(0) = C'(0) = 0$ and $C''(\cdot) > 0$.^{10,11} Therefore, with the possibility of intervention, the payoffs of the principal and the agent are $U_P(x - \Delta, \theta) - \delta C(|\Delta|)$ and $U_A(x - \Delta, \theta; b)$, respectively. Depending on the application of the model, the parameter $\delta > 0$ captures the principal's aversion to confrontation, her busyness and alternative use of time and resources (i.e., other tasks she is overseeing), the culture and bureaucracy of the organization, or the complexity and scope of the underlying task. Since intervention is more costly than communication, it is natural to assume that the principal intervenes only after communicating with the agent (Simon (1947)). Finally, payoffs are realized and distributed to the principal and the agent.

B. Equilibrium with Communication and Intervention

I solve for Perfect Bayesian Equilibria of the game in pure strategies.¹² The formal definition is given in the Appendix. I begin the analysis with the principal's intervention. The principal intervenes whenever the agent's choice deviates from her ideal project, that is, $x \neq \theta$. The agent chooses $x \neq \theta$ either because he is uninformed about θ or because he is biased. Let $\Delta^*(x, \theta)$ be the

¹⁰ I also assume that $\lim_{\Delta \rightarrow \infty} C''(\Delta) > 0$, which is invoked in the proof of Lemma 1. Without this assumption, the agent behaves as if $\beta = \infty$ but the main result continues to hold. In addition, I assume that $\lim_{\Delta \rightarrow \infty} \frac{\partial^\kappa C(\Delta)}{\partial \Delta^\kappa}$ is finite for some integer $\kappa \geq 2$ (κ can be arbitrarily large). This assumption is invoked in the proof of Proposition 1, part (i). Note that both assumptions hold for quadratic cost functions.

¹¹ The Internet Appendix, Section I.E, shows that intervention also hinders communication if the intervention cost is fixed.

¹² Assuming that the principal plays pure strategies is without loss of generality. However, allowing the agent to play mixed strategies could potentially add equilibria unless one also assumes that the agent's indirect utility function $V_A(x, \theta; b)$ is concave in x . Concavity is guaranteed if, for example, $C'''(\cdot) \leq 0 \leq L'''(\cdot)$.

principal's optimal intervention policy conditional on project x and state θ . Regardless of the message the principal sent beforehand, $\Delta^*(x, \theta)$ solves

$$\max_{\Delta} U_P(\theta, \theta) - L(|x - \theta - \Delta|) - \delta C(|\Delta|) \quad (3)$$

and is given by the unique solution of

$$L'(|x - \theta - \Delta|) = \delta C'(|\Delta|). \quad (4)$$

Notice that the intervention function depends on x and θ only through $x - \theta$, and it has several intuitive properties. First, since the principal has stronger incentives to intervene when the agent's decision is more detrimental, $|\Delta^*(x - \theta)|$ increases in $|x - \theta|$. Second, the principal's intervention reduces the distance between the implemented project and θ , that is, $|x - \Delta^*(x - \theta) - \theta| < |x - \theta|$. Finally, since the principal has stronger incentives to intervene when it is less costly to do so, $|\Delta^*(x - \theta)|$ decreases in δ .¹³

To illustrate the main results of the paper, I develop an example using quadratic utility and cost functions throughout the analysis.¹⁴

EXAMPLE 1: Suppose $U_P = A - (x - \theta)^2$, $U_A = A - (x - \theta - b)^2$, and $C = \Delta^2$. Then, the principal's optimal intervention policy is $\Delta^*(x - \theta) = \frac{x - \theta}{1 + \delta}$.

The principal and the agent have rational expectations about $\Delta^*(x - \theta)$, which they both take into account at the communication stage. Specifically, the principal maximizes

$$V_P(x, \theta) \equiv U_P(x - \Delta^*(x - \theta), \theta) - \delta C(|\Delta^*(x - \theta)|). \quad (5)$$

Notice that the principal's ideal point remains θ even though she has the option to intervene. Intuitively, a smaller $|x - \theta|$ not only increases the principal's direct utility, but also reduces her need to intervene, which is costly. As a result, in equilibrium the principal will send the message that induces the agent to choose the project that is closest to θ among all projects on the equilibrium path.

Similarly, given message m , the agent chooses the project that maximizes

$$V_A(x, \theta; b) \equiv U_A(x - \Delta^*(x - \theta), \theta; b). \quad (6)$$

Recall that the agent's utility is maximized when the project that is implemented following the principal's intervention is $\theta + b$. Therefore, the agent has incentives to choose an initial project x_A that satisfies

$$x_A - \Delta^*(x_A - \theta) = \theta + b. \quad (7)$$

¹³ Notice that the principal cannot commit to policies that are ex post suboptimal even if they are ex ante beneficial (e.g., $\Delta \equiv x - \theta$). Partial commitment would be equivalent to assuming that the principal's cost of intervention is lower, but the main results continue to hold as long as the principal has limited commitment.

¹⁴ All formal derivations of the quadratic utility and cost functions example are given in Section I.F of the Internet Appendix.

Since $\Delta^*(x_A - \theta)$ is the solution of equation (4) and project x_A satisfies equation (7), it follows that $L'(b) = \delta C'(x_A - \theta - b)$. Rewriting this equality gives $x_A = \theta + \beta$, where

$$\beta \equiv b + (C')^{-1} \left(\frac{L'(b)}{\delta} \right). \quad (8)$$

Therefore, unlike the principal, intervention changes the agent's ideal point.

LEMMA 1: *The agent's indirect utility $V_A(x, \theta; b)$ obtains its unique maximum at $x = \theta + \beta$.*

Lemma 1 implies that the agent behaves as if his bias is β rather than b . Therefore, β can be interpreted as the agent's intervention-induced bias. Substituting x_A with $\theta + \beta$ in equation (7) implies that β can also be expressed as

$$\beta = b + \Delta^*(\beta). \quad (9)$$

The agent's intervention-induced bias is equal to his intrinsic bias plus the amount of intervention that is expected by the principal if the agent were to choose his “new” ideal point, $\theta + \beta$. In other words, the agent's new ideal point “overshoots” relative to the project that is eventually implemented by the principal. It is straightforward to see that the extent of overshooting, which is captured by $\Delta^*(\beta) = (C')^{-1}(\frac{L'(\beta)}{\delta})$, is decreasing in δ and increasing in b . Intuitively, the agent expects less intervention by the principal when the cost of intervention is higher, and thus less overshooting is needed to guarantee that the final project would be closer to $\theta + b$. Similarly, a larger b implies that the agent's ideal point is more distant from the principal's ideal point. Since the principal has stronger incentives to intervene when $|x - \theta|$ is larger, the agent expects more intervention when his bias is larger, which implies that more overshooting is needed to guarantee $\theta + b$ as the final project.

EXAMPLE 2: Under quadratic utility and cost functions, the indirect utilities of the principal and the agent are $V_P = A - \frac{\delta}{1+\delta}(x - \theta)^2$ and $V_A = A - \frac{\delta^2}{(1+\delta)^2}(x - \theta - \beta)^2$, respectively, where the agent's intervention-induced bias is $\beta = b + \delta^{-1}b$.

Ultimately, the quality of communication between the principal and the agent is determined by their conflict of interests as reflected by the differences in their indirect utility functions. Similar to Crawford and Sobel (1982), communication in equilibrium is characterized by a partition $(a_0, a_1, \dots, a_n)$ of $[\underline{\theta}, \bar{\theta}]$. In equilibrium, the principal's message is noisy—it reveals the partition element to which the realized state belongs, but not the actual state. If the principal's message indicates $\theta \in (a_{i-1}, a_i)$, then the agent chooses project x_i^* that maximizes $E[V_A(x, \theta; b) | \theta \in (a_{i-1}, a_i)]$. Since the agent is biased toward overinvestment (i.e., $b > 0$) and intervention amplifies this bias (i.e., $\beta > b$),

$x_i^* > \frac{a_{i-1}+a_i}{2} + b$,¹⁵ that is, the agent chooses a project that is strictly larger than the conditional expected value of θ plus his intrinsic bias. The optimality of the principal's communication strategy requires her to be indifferent between two adjacent actions when θ falls on the boundary between the partition elements that induce them. In other words, a_i has to be the midpoint between x_i^* and x_{i+1}^* . Together with the agent's best response, these observations imply

$$a_{i+1} - a_i < a_i - a_{i-1} - 4b. \quad (10)$$

According to the inequality in condition (10), the size of a partition element is at least $4b$ smaller than the preceding one. Intuitively, the principal has incentives to understate the benefit from large projects to undo the bias of the agent. As a result, the principal has less credibility when her message states that θ is small. The reduced credibility is reflected by a larger interval, which means that less information is communicated by the principal. The constraint on the relative size of the partition elements restricts their number in equilibrium to be smaller than a positive integer, which I denote by N_{in}^* . An informative equilibrium exists as long as $N_{in}^* \geq 2$.¹⁶

In general, a larger N_{in}^* implies that the partition is finer, and hence, more information is revealed in equilibrium by the message of the principal. The quality of communication in equilibrium can also be measured by the residual uncertainty of θ following the principal's message. Specifically, the residual variance of θ in the most informative equilibrium is defined as

$$\Sigma_{in}^* \equiv E[(E[\theta|m^*(\theta)] - \theta)^2], \quad (11)$$

where $m^*(\theta)$ is the principal's message in this equilibrium when the realized state is θ .

EXAMPLE 3: Consider the equilibrium under quadratic utility and cost functions. The agent's optimal choice is $x_i^* = \frac{a_{i-1}+a_i}{2} + \beta$ and the partition satisfies $a_{i+1} - a_i = a_i - a_{i-1} - 4\beta$. The largest number of elements a partition can have is $N_{in}^* = N(\beta)$, where

$$N(\beta) \equiv \left\lceil -\frac{1}{2} + \frac{1}{2} \sqrt{1 + 2 \frac{\bar{\theta} - \theta}{\beta}} \right\rceil. \quad (12)$$

The residual variance of θ is $\Sigma_{in}^* = \Sigma(\beta)$, where

$$\Sigma(\beta) \equiv \frac{\sigma^2}{N(\beta)^2} + \frac{\beta^2(N(\beta)^2 - 1)}{3} \quad (13)$$

and $\sigma^2 \equiv \frac{(\bar{\theta} - \theta)^2}{12}$ is the unconditional variance of θ .

¹⁵ The proof of Theorem 1 shows that x_i^* is the unique solution of $x = \frac{a_{i-1}+a_i}{2} + b + \frac{1}{2}[\Delta^*(x - a_{i-1}) + \Delta^*(x - a_i)]$.

¹⁶ As in Crawford and Sobel (1982), for every $n \in \{1, \dots, N_{in}^*\}$ there is an equilibrium in which the partition has n elements.

¹⁷ The notation $\lceil r \rceil$ is used to indicate the smallest integer greater than or equal to r .

C. Does Intervention Hinder Communication?

Intervention hinders communication if the quality of communication in equilibrium with intervention is lower than it is without it. The communication game without intervention, which was studied by Crawford and Sobel (1982), is a special case of our analysis in which intervention is prohibitively costly (i.e., $\delta = \infty$, which implies $\Delta^*(x - \theta) \equiv 0$). Specifically, in equilibrium without intervention, the agent's optimal choice is $x_i^* = \frac{a_{i-1} + a_i}{2} + b$, the partition satisfies condition (10) with equality, $N_{no}^* = N(b)$, and $\Sigma_{no}^* = \Sigma(b)$.¹⁸ Therefore, the effect of intervention on communication is captured by comparing $\{N_{in}^*, \Sigma_{in}^*\}$ with $\{N_{no}^*, \Sigma_{no}^*\}$, the analogs for the communication game without intervention.¹⁹

In general, the quality of communication between the principal and the agent deteriorates as their preferences become more misaligned. Since intervention amplifies the conflict between the principal and the agent (i.e., $\beta > b$), intervention hinders communication.

THEOREM 1:

- (i) *The largest partition in equilibrium with intervention is coarser than the largest partition in equilibrium without intervention, that is, $N_{in}^* \leq N_{no}^*$.*
- (ii) *The residual uncertainty in the most informative equilibrium with intervention is larger than the residual uncertainty in the most informative equilibrium without intervention, with the inequality being strict if and only if the most informative equilibrium without intervention is not a babbling equilibrium, that is, if $N_{no}^* = 1$, then $\Sigma_{in}^* = \Sigma_{no}^*$, and if $N_{no}^* \geq 2$, then $\Sigma_{in}^* > \Sigma_{no}^*$.*

Theorem 1 is the main result of the paper. Part (i) shows that the largest partition in equilibrium with intervention has fewer elements than the largest partition in equilibrium without intervention. Indeed, since with intervention the difference between each consecutive interval is strictly larger than $4b$, the largest number of intervals that spans $[\underline{\theta}, \bar{\theta}]$ must be smaller with intervention than it is without it. Moreover, since the size of the intervals decreases at a higher rate, the intervals are more uneven in a game with intervention, which implies that the residual variance is larger. Indeed, part (ii) shows that as long as some information can be revealed in equilibrium without intervention, the possibility of intervention strictly reduces the amount of information that can be revealed in equilibrium by the principal. Moreover, if no information can be revealed in equilibrium without intervention, then any equilibrium with intervention is also uninformative.

¹⁸ The characterization of the equilibrium without intervention does not assume quadratic utility functions.

¹⁹ The comparison of the quality of communication in Theorem 1 is determined by the properties of the most informative equilibrium in both setups. Focusing on the most informative equilibrium is standard in cheap-talk games. Indeed, Crawford and Sobel (1982) show that the most informative equilibrium (Pareto) dominates any other equilibrium. See also Chen, Kartik, and Sobel (2008) for an alternative justification for the selection of the most informative equilibrium in cheap-talk games.

The intuition is as follows. Recall that in anticipation of intervention, the agent chooses a project that is larger than his best estimate of $\theta + b$. By overshooting, the agent increases the cost that the principal has to incur to downsize the project and bring it closer to θ . Because the agent is expected to overshoot, the principal has even stronger incentives to understate the true value of θ in an effort to counter the excessive investment. The agent, however, understands the motives of the principal to prevent overshooting and in response puts even less weight on the credibility of her messages. As a result, the ability of the principal to influence the agent's decision by communicating her private information is reduced. Taken together, less information is revealed in equilibrium, that is, intervention hinders communication. The next example illustrates this dynamic.

EXAMPLE 4: Suppose the utility and cost functions are quadratic with $b = \frac{1}{6}$, $\delta = 1$, and $[\underline{\theta}, \bar{\theta}] = [0, 1]$. I proceed in four steps:

- (i) First, consider the game without intervention. Since condition (10) holds with equality, its solution in this example implies that the most informative equilibrium features a partition with two elements (i.e., $N_{no}^* = 2$), where the cutoff is $\theta^* = \frac{10}{12}$, that is, the principal reveals the location of θ with respect to $\frac{10}{12}$, but nothing else. If the principal reveals $\theta \in [0, \frac{10}{12}]$, then the agent updates his beliefs about the expected value of θ from $\frac{1}{2}$ to $\frac{5}{12}$, and due to his bias b , he chooses $x_1 = \frac{5}{12} + \frac{1}{6} = \frac{7}{12}$. Similarly, if the principal reveals $\theta \in [\frac{10}{12}, 1]$, then the agent chooses $x_2 = \frac{11}{12} + \frac{1}{6} = \frac{13}{12}$. Since $\theta^* = \frac{10}{12}$ is the midpoint between $x_1 = \frac{7}{12}$ and $x_2 = \frac{13}{12}$, the principal's communication strategy is indeed optimal. These observations are summarized in panel A of Figure 1.
- (ii) How does intervention affect this equilibrium? If the agent chooses project x , then the principal's intervention is $\Delta^* = \frac{x-\theta}{2}$, and the final project is $x - \Delta^* = \frac{1}{2}[x + \theta]$, that is, the principal intervenes such that the final project is the midpoint between the agent's initial choice and the true state. As a result, if the agent does not adjust his decisions, then he can expect the final project to be $\frac{1}{2}[\frac{7}{12} + \frac{5}{12}] = \frac{6}{12}$ when $\theta \in [0, \frac{10}{12}]$ and $\frac{1}{2}[\frac{13}{12} + \frac{11}{12}] = \frac{12}{12}$ otherwise. On average, the principal downsizes projects by $\frac{1}{12}$. The effect of intervention on the final project is illustrated in panel B of Figure 1.
- (iii) To ensure that the final projects remain $\frac{7}{12}$ when $\theta \in [0, \frac{10}{12}]$ and $\frac{13}{12}$ otherwise, the agent must increase their initial size to $x'_1 = \frac{9}{12}$ and $x'_2 = \frac{15}{12}$, respectively. Indeed, since $x - \Delta^* = \frac{1}{2}[x + \theta]$, x'_1 must satisfy $x_1 = \frac{1}{2}[x'_1 + \frac{5}{12}]$ and x'_2 must satisfy $x_2 = \frac{1}{2}[x'_2 + \frac{11}{12}]$. The overshooting by the agent is captured in panel C of Figure 1. Since these projects are larger (by $\frac{1}{3}$) than the conditional expected value of θ in the corresponding intervals, the agent behaves as if his bias is $\frac{1}{3}$ rather than $\frac{1}{6}$, that is, the intervention-induced bias is $\beta = \frac{1}{3}$.

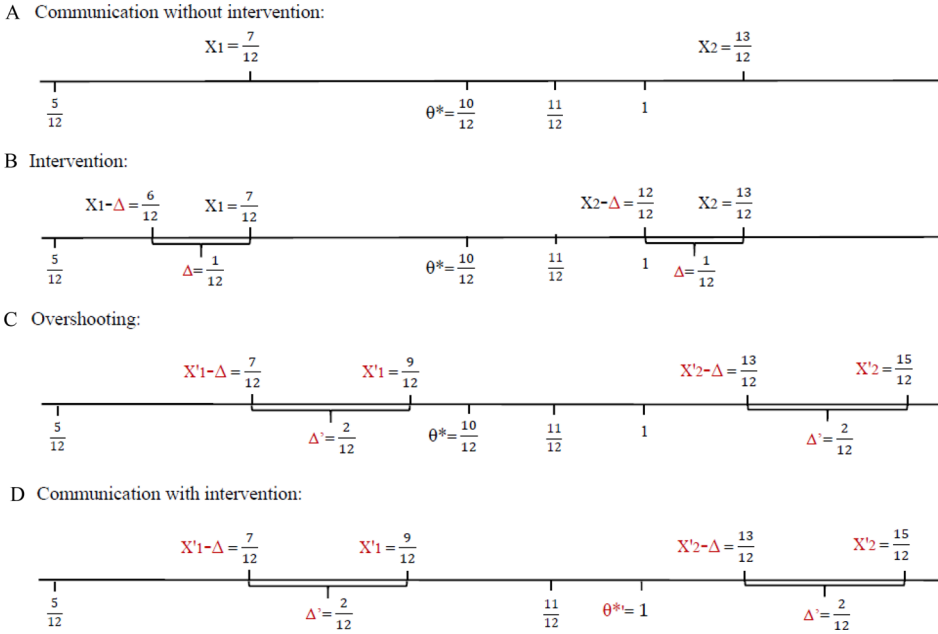


Figure 1. Example 4 with quadratic utility and cost functions, $b = \frac{1}{6}$, $\delta = 1$, and $[\theta, \bar{\theta}] = [0, 1]$. (Color figure can be viewed at wileyonlinelibrary.com)

- (iv) Finally, the principal, who anticipates the overshooting by the agent, prefers project $x'_1 = \frac{9}{12}$ over $x'_2 = \frac{15}{12}$ if and only if $\theta < \theta^{*'} = 1$. The shift of the communication cutoff from $\theta^* = \frac{10}{12}$ to $\theta^{*'} = 1$ is depicted in panel D of Figure 1. Since $\bar{\theta} = 1$, the principal prefers the smaller project for any realization of θ . Therefore, she cannot resist the temptation to understate the true value of θ by pretending that $\theta \in [0, \frac{10}{12}]$ even if $\theta \in [\frac{10}{12}, 1]$. In other words, the message of the principal is uninformative about θ , and as a result it is ignored by the agent. The only equilibrium with intervention is an uninformative equilibrium, that is, $N_{in}^* = 1$. Since $N_{no}^* = 2$, $N_{in}^* < N_{no}^*$ and intervention hinders communication.

The discussion above shows that intervention hinders communication by providing the agent with perverse incentives to overshoot. However, intervention also hinders communication by providing the agent with an informational benefit.²⁰

²⁰ The informational benefit of intervention exists even if the agent cannot overshoot. In the Internet Appendix, Section I.A, I consider a binary version of the model. With only two feasible projects, the agent cannot overshoot—he can either comply or disobey the principal's request. The key difference is that with only two options, the agent takes more risk not by choosing larger projects, but rather by disobeying the principal more frequently when the latter recommends the smaller project. The main result continues to hold.

To understand this observation, recall that the principal's message in equilibrium reveals that θ is in the interval (a_{i-1}, a_i) , but does not reveal its exact location therein. Noisy communication is the inevitable outcome of the principal's desire to prevent overinvestment. Since the agent remains uncertain about the value of θ , his optimal choice in equilibrium trades off the risk of overshooting his own ideal point with the risk of undershooting it. Nevertheless, the bias of the agent toward overinvestment ($b > 0$) implies that his optimal action x_i^* is closer to a_i than it is to a_{i-1} . Indeed, $x_i^* > \frac{a_{i-1} + a_i}{2}$. Since $|\Delta^*(x - \theta)|$ is increasing in $|x - \theta|$, on average the principal intervenes more aggressively when $\theta \in (a_{i-1}, x_i^*)$ than when $\theta \in (x_i^*, a_i)$. In particular, the principal is more likely to downsize the project than she is to increase its size. This asymmetry reveals additional information about the location of θ in (a_{i-1}, a_i) . It also implies that the agent faces a lower risk of overshooting his own ideal point. On the margin, knowing that the principal will intervene more aggressively when x overshoots θ than when it undershoots θ emboldens the agent to take more "risk" by choosing even larger projects. By choosing the project closer to a_i , the agent provokes more intervention and as a result is able to elicit more information by conditioning on the principal's decision.²¹ Since the agent has an alternative source of information, he behaves as if the uncertainty about θ is smaller, which is reflected by a larger bias toward overinvestment. Overall, the agent is less receptive to the principal's messages and less information is revealed in equilibrium.

EXAMPLE 5: Consider Example 4. Recall that without intervention the agent's utility is $A - (x - \theta - \frac{1}{6})^2$. If the principal has the power to intervene, x is downsized by $\Delta^* = \frac{x - \theta}{2}$. Therefore, with intervention, the agent's indirect utility is $A - (x - \theta - \frac{x - \theta}{2} - \frac{1}{6})^2 = A - (\frac{x - \theta}{2} - \frac{1}{6})^2$. Since the term $x - \theta$ is now scaled by a factor of $\frac{1}{2}$, the agent effectively puts less weight on the value of calibrating the project to the fundamentals of the firm when responding to the principal's message; the agent can rely more heavily on the principal's informed intervention as a substitute. Indeed, the possibility of intervention assures the agent that x cannot be too far away from θ . This is the informational benefit of intervention. Equivalently, the agent's indirect utility can be expressed as $A - \frac{1}{4}(x - \theta - \frac{1}{3})^2$, which implies that the agent behaves as if his bias is $\beta = \frac{1}{3}$ rather than $b = \frac{1}{6}$. In other words, the behavior of an agent who puts less weight on the fundamentals of the firm is equivalent to the behavior of an agent who puts more weight on firm fundamentals but has a larger bias.

D. Welfare Implications

This subsection studies the welfare implications of the model. For this purpose, I define V_P^{**} and U_P^{**} as the principal's expected utility in the most

²¹ Notice that the agent cannot revise his decision after this new information is revealed, but he can condition his decision on the information that is embedded in the principal's intervention. This reasoning is similar to the effect of the winner curse in common value auctions and pivotal considerations in games with strategic voting.

informative equilibrium, with and without intervention, respectively. I similarly define V_A^{**} and U_A^{**} for the agent. Recall that in a cheap-talk model, there always exists an uninformative babbling equilibrium, which is equivalent to an equilibrium without communication. To study how communication itself affects the welfare implications of intervention, I also define V_P^* and U_P^* as the principal's expected utility in the babbling equilibrium, with and without intervention, respectively.

D.1. Does Intervention Benefit the Principal?

Intervention has several opposing effects on the principal. On the one hand, intervention is a correction device—the principal can use her information on θ to undo the agent's bias and choose the desired project. On the other hand, with intervention, the agent is more likely to overinvest and less information can be communicated in equilibrium. Both forces reduce the principal's welfare since the agent is less likely to choose the desired project. The next result provides sufficient conditions under which these adverse effects dominate in equilibrium.

PROPOSITION 1:

- (i) *There exist $\bar{\delta} > 0$ and $\bar{b} > 0$ such that, all else equal, if $\delta < \bar{\delta}$ or $b > \bar{b}$, then $V_P^{**} < U_P^{**}$.*
- (ii) *There exist $\underline{b} > 0$ and $\underline{\delta} \in (0, \bar{\delta})$ such that if $b < \underline{b}$ and $\delta \in (\underline{\delta}, \bar{\delta})$, then $N_{no}^* \geq 2$, $N_{in}^* = 1$, and $V_P^* > U_P^*$.*

Part (i) of Proposition 1 has the following striking implication: if the cost of intervention is sufficiently small ($\delta < \bar{\delta}$) or the underlying agency problem is severe ($b > \bar{b}$), the principal is better-off without the power to intervene in the agent's decision ($V_P^{**} < U_P^{**}$). In other words, from the principal's perspective intervention is least desirable when it is seemingly most effective or most needed. Intuitively, with intervention, the agent overshoots just enough to ensure that the final project is the same as would obtain without intervention. Since the principal is expected to intervene more aggressively when the cost of intervention is small or when overinvestment is significant, the distortion in the agent's incentives is particularly large when δ is small or b is large (i.e., β decreases with δ and increases with b). As β increases, the first-order concern of the principal is not to calibrate the project to θ , but rather to undo the tendency of the agent to overshoot as much as possible. In other words, the benefit of intervention as a correction device that allows the principal to use her private information becomes of second order. In equilibrium, the principal ends up with the same level of overinvestment as would be obtained without intervention, but now she also pays for the cost of intervention and communication is ineffective. As a result, the principal is better-off without intervention.

Part (ii) tightens the conditions of part (i) to ensure that the principal is better-off without intervention solely because of its adverse effect on communication. Specifically, the requirements $\delta \in (\underline{\delta}, \bar{\delta})$ and $b < \underline{b}$ ensure that an

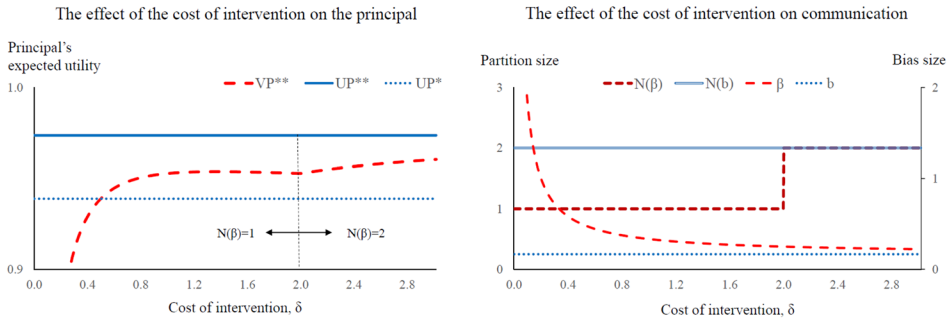


Figure 2. The principal's expected utility in equilibrium when $A = 1$, $\bar{\theta} - \underline{\theta} = 1$, and $b = \frac{1}{6}$. (Color figure can be viewed at wileyonlinelibrary.com)

informative equilibrium exists in a game without intervention (i.e., b is small enough to ensure that $N_{no}^* \geq 2$), but does not exist in a game with intervention (i.e., δ is small enough to ensure that $N_{in}^* = 1$), that is, intervention strictly hinders communication. Moreover, these additional requirements ensure that in the absence of communication, the benefit from informed intervention dominates the distortion it induces in the agent's incentives (i.e., δ cannot be too small, to ensure that $V_P^* > U_P^*$). Together with part (i), this result implies that intervention reduces the principal's welfare because it hinders communication.

EXAMPLE 6: Under quadratic utility and cost functions, we have $U_P^* = A - (\sigma^2 + b^2)$, $U_P^{**} = A - (\Sigma(b) + b^2)$, $V_P^* = A - \frac{\delta}{1+\delta}(\sigma^2 + \beta^2)$, and $V_P^{**} = A - \frac{\delta}{1+\delta}(\Sigma(\beta) + \beta^2)$. Figure 2 illustrates Proposition 1 when $N(b) = 2$. The right panel shows that $N(\beta) = 1$ if and only if $\delta < 2$. It also plots β and b as a function of δ . The left panel shows that $V_P^{**} < U_P^{**}$ and $U_P^* < V_P^{**}$ if and only if $\delta > 0.5$. Taken together, if $0.5 < \delta < 2$, then intervention harms the principal (since $V_P^{**} < U_P^{**}$), but only because it hinders communication (since $N(\beta) < N(b)$ and $V_P^* > U_P^*$).²²

D.2. Does Intervention Benefit the Agent?

On the surface, intervention may be expected to harm the agent as it not only hinders communication but also allows the principal to override the agent's decision. The next result shows, however, that the agent actually benefits from intervention, especially when the cost of intervention to the principal is small or the underlying agency problem is severe.

PROPOSITION 2: *There exist $\bar{\delta} > 0$ and $\bar{b} > 0$ such that, all else equal, if $\delta < \bar{\delta}$ or $b > \bar{b}$, then $V_A^{**} > U_A^{**}$.*

How can the agent benefit from the principal's intervention? Notice that intervention is an informed decision, and as explained in Section I.C, it is

²² Notice that if $\delta < 2$, then $N(\beta) = 1$, that is, the most informative equilibrium with intervention is the babbling equilibrium. Therefore, $\delta < 2$ implies $V_P^* = V_P^{**}$.

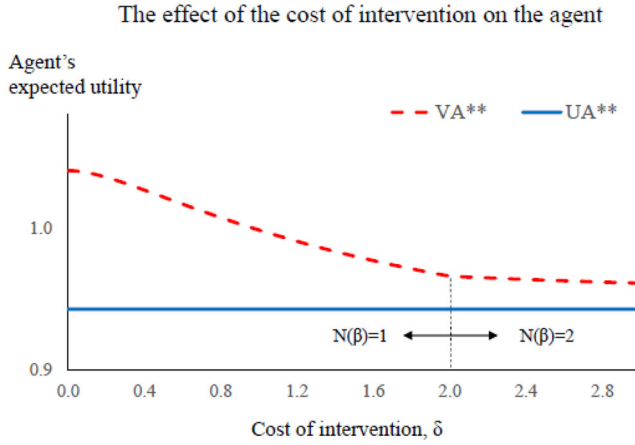


Figure 3. The agent's expected utility in equilibrium when $A = 1$, $\bar{\theta} - \underline{\theta} = 1$, and $b = \frac{1}{6}$. (Color figure can be viewed at wileyonlinelibrary.com)

an alternative way through which the agent can elicit information from the principal. When δ is small or b is large, the intervention policy of the principal is relatively aggressive and as such is very sensitive to the principal's private information. By choosing the “correct” amount of overshooting, the agent can both use the information in the principal's decision to intervene and at the same time ensure that the final project is closer to his ideal point. The informational benefit of intervention is large enough to fully compensate the agent for the loss of information through communication.

EXAMPLE 7: Under quadratic utility and cost functions, we have $U_A^{**} = A - \Sigma(b)$ and $V_A^{**} = A - \frac{\delta^2}{(1+\delta)^2} \Sigma(\beta)$. Figure 3 illustrates Proposition 2.

REMARK 1: Although both the principal and the agent benefit from more effective communication (Crawford and Sobel (1982)), the contrast between Propositions 1 and 2 demonstrates that the two can have opposite preferences with respect to intervention.

II. Extensions

A. Pay For Performance

The incompleteness of contracts plays a central role in the analysis as actions and messages cannot be contracted on. However, the value of the project (e.g., its terminal cash-flows) could in principle be contracted on. For example, the principal could offer the agent a fraction $\omega \in (0, 1)$ of $U_P(x, \theta)$, the value of the project.

To illustrate the effect of ω on the analysis, suppose that the agent's intrinsic private benefit from investment is Bx , where $B > 0$. Given ω , the principal's utility is $(1 - \omega)U_P(x, \theta)$ and the agent's utility is $Bx + \omega U_P(x, \theta)$. Notice that

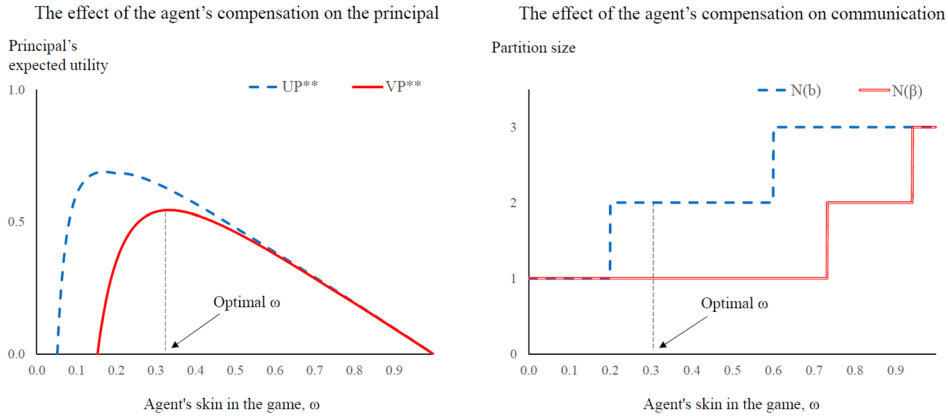


Figure 4. The principal's expected utility as a function of ω when $A = 1$, $\bar{\theta} - \underline{\theta} = 1$, $B = 0.1$, and $\delta = 0.1$. (Color figure can be viewed at wileyonlinelibrary.com)

the latter obtains its unique maximum at $\theta + b(\omega)$, where $b(\omega) \equiv (L')^{-1}(\frac{B}{\omega})$. As in the baseline model, the agent's intervention-induced bias is given by equation (8), with the exception that b is replaced by $b(\omega)$ and δ is replaced by $\frac{\delta}{1-\omega}$. Specifically,

$$\beta(\omega) \equiv (L')^{-1}\left(\frac{B}{\omega}\right) + (C')^{-1}\left(\frac{B}{\delta} \frac{1-\omega}{\omega}\right). \quad (14)$$

Note that $\beta(\omega) > b(\omega)$, which is a decreasing function of ω . Intuitively, with a larger ω , the agent internalizes more of the benefit from choosing a project of size θ and as a result his bias is smaller. Moreover, a larger ω implies that the principal puts less weight on maximizing $U_P(x, \theta)$ and more weight on the cost of intervention. In effect, the cost of intervention per unit of utility is larger. As discussed in Section I, β increases in b and decreases in δ . Since $b(\omega)$ decreases in ω and $\frac{\delta}{1-\omega}$ increases in ω , the agent's intervention-induced bias decreases with ω , although it is always larger than $b(\omega)$.

When choosing ω , the principal trades off the direct cost of giving away part of the project's value to the agent with the benefit of lowering $\beta(\omega)$. In general, the ex ante optimal level of ω may depend on factors that are outside of the model (e.g., the agent's ability). Nevertheless, the example below demonstrates that the principal can be better-off without intervention even if ω is chosen optimally.

EXAMPLE 8: Under quadratic utility and cost functions, we have $b(\omega) = \frac{B}{2\omega}$, $\beta(\omega) = b(\omega) + b(\omega)\frac{1-\omega}{\delta}$, $U_P^{**} = (1-\omega)(A - (\Sigma(b(\omega)) + b(\omega)^2))$, and $V_P^{**} = (1-\omega)(A - \frac{\delta}{1-\omega+\delta}(\Sigma(\beta(\omega)) + \beta(\omega)^2))$. The left panel of Figure 4 shows that the optimal ω is $\frac{1}{3}$, and if $\omega = \frac{1}{3}$, then $V_P^{**} < U_P^{**}$. The right panel shows that $N(b(\omega)) = 2$ and $N(\beta(\omega)) = 1$ when $\omega = \frac{1}{3}$. Therefore, intervention hinders communication and harms the principal's wealth even if the principal chooses ω optimally.

B. Agent's Intervention Cost

In the baseline model, intervention does not impose a direct cost on the agent. In some cases, however, intervention may have a direct negative effect on the agent's reputation, ego, or compensation. To consider this possibility, suppose that intervention imposes a cost $\tau K(|\Delta|)$ on the agent, where $K''(\cdot) > 0$, $K(0) = K'(0) = 0$, and $\tau > 0$. The baseline model assumes $\tau = 0$. In the Internet Appendix, Section I.B, I show that the agent's intervention-induced bias can be generalized to

$$\beta = \pi + (C')^{-1} \left(\frac{L'(\pi)}{\delta} \right), \quad (15)$$

where $\pi \in (0, b]$ is the solution of

$$T'(b - \pi) = L''(\pi) \frac{\tau K' \left((C')^{-1} \left(\frac{L'(\pi)}{\delta} \right) \right)}{\delta C'' \left((C')^{-1} \left(\frac{L'(\pi)}{\delta} \right) \right)}. \quad (16)$$

Moreover, I show that β decreases with τ and there exists $\tau^* > 0$ such that $\beta > b$ if and only if $\tau < \tau^*$. For example, under the quadratic functional form (i.e., $K(|\Delta|) = \Delta^2$),

$$\beta = b \frac{1 + \delta}{\tau/\delta + \delta} \quad (17)$$

and $\beta > b$ if and only if $\tau < \delta$, that is, the agent's intervention-induced bias is greater than his intrinsic bias whenever the intervention cost for the agent is smaller than the intervention cost for the principal.

In general, if $\tau < \tau^*$, then $\beta > b$ and intervention hinders communication for the same reasons as in the baseline model. However, if $\tau > \tau^*$, then $\beta < b$ and intervention facilitates communication. Intuitively, the best way to avoid the unpleasant consequences of intervention is to follow the principal's instructions. When τ is large, overshooting becomes costly to the agent since the principal is more likely to intervene when the agent chooses a project that is further away from θ . Since the agent is less likely to overshoot, the principal has fewer incentives to understate θ , and more information can be revealed in equilibrium when τ is large.²³

REMARK 2: *If intervention facilitates communication, then it also increases the principal's expected utility in equilibrium. Intuitively, when $\tau > 0$, the fear of intervention discourages the agent from overinvestment and increases the revelation of information. Both forces benefit the principal. This observation*

²³ This prediction differs from that of Adams and Ferreira (2007), who argue (in the context of corporate boards) that communication is ineffective when the agent's disutility from being monitored is high. In their model, the agent has fewer incentives to cooperate with the principal when monitoring induces larger costs and, by assumption, without cooperation the principal cannot advise the agent.

also implies that if $\tau > 0$ and δ is sufficiently small, then $\beta < b$ and the principal benefits from the power to intervene.²⁴

C. Informed Agent

In some applications of the model, the agent may also have private information about θ . To consider this possibility, suppose that $\theta = \theta_P + \theta_A$, where θ_P is independent of θ_A . I assume that the principal is privately informed about θ_P and the agent is privately informed about θ_A .

Harris and Raviv (2005) study a version of this model without intervention. They show that the set of equilibria with a privately informed agent is equivalent to the set of equilibria with an uninformed agent, with the exception that $x^*(\theta_A, m) = \theta_A + x^*(m)$, where $x^*(m)$ is the project that the uninformed agent would have chosen in equilibrium. Intuitively, the informed agent's optimal decision fully incorporates his private information about θ_A in a way that leaves the principal's expected utility independent of the realization of θ_A . Therefore, the quality of communication is unaffected by the agent's private information.

In a model with intervention, the principal may change her intervention policy based on what she learns about θ_A from the agent's decision. The ability to signal private information could change the informed agent's initial decision, which in turn might affect the principal's ability to influence the agent in the first place. Nevertheless, in the Internet Appendix, Section I.C, I show that under quadratic utility and cost functions, the informed agent's intervention-induced bias is $b + \delta^{-1}b$, which is the same as the uninformed agent's intervention-induced bias. This result demonstrates that intervention can also hinder communication in a setup with two-sided information asymmetries.

Intuitively, the agent's decision in equilibrium moves one-to-one with the value of θ_A . Since a larger project signals a higher value of θ_A , the principal infers that her ideal point also shifts upward. Importantly, the distance between the agent's choice and the principal's ideal point is determined solely by the variation in θ_P . Since the distortion in the agent's decision and the intensity of the principal's intervention are unaffected by θ_A , the analysis of the baseline model holds.²⁵

REMARK 3: *The principal's expected utility under the most informative equilibrium is identical to her expected utility when θ_A is common knowledge. This*

²⁴ If $\tau > 0$, then the welfare results in Section I.D.1 (with respect to δ) require that δ be in an intermediate range to ensure $\beta > b$.

²⁵ Interestingly, in the Internet Appendix, Section I.C, I also show that if $\tau > 0$, then the informed agent's intervention-induced bias is strictly larger than $b \frac{1+\delta}{\tau/\delta+\delta}$. Recall from Section II.B that the intervention-induced bias under quadratic utility and cost functions is exactly $\frac{1+\delta}{\tau/\delta+\delta}$ when the agent is uninformed. Therefore, at least in this setup, the agent's private information exacerbates the adverse effect of intervention on communication. In this respect, this analysis predicts that communication is less effective when the agent is better informed.

result follows from the observation that the agent's decision in equilibrium fully incorporates his private information about θ_A ; the only loss of welfare stems from the noisy communication of θ_P . Therefore, under quadratic functional forms, the principal's expected utility can be calculated using the expressions from Example 6 in Section I.D.1. In particular, the principal can also be better-off without intervention also when the agent is informed.

III. Applications and Empirical Implications

The above model has two main predictions. First, it predicts that the frequency with which control rights are exercised should be negatively correlated with the quality and prevalence of communications. The negative correlation between these two endogenous variables is intuitive: if communication is more effective, then the agent's choice of projects in equilibrium is closer to the principal's ideal point θ , and on average the principal has less need to intervene. Formally, in the Internet Appendix, Section I.D, I show that the expected intervention in equilibrium with effective communication is strictly smaller than in equilibrium without communication. Second, the model predicts that the quality of communication increases with the cost of intervention (i.e., β decreases with δ). As discussed in the setup of the model, the cost of intervention is related to various characteristics of the principal, the organization, and the underlying task. Therefore, cross-sectional variation in factors that increase the cost of intervention should be positively correlated with the prevalence and effectiveness of communication.

Below, I discuss four applications of the model: managerial leadership, corporate boards, PE, and shareholder activism. In all of these applications, there exist significant information asymmetries and conflicts of interest that cannot be easily removed by contracts. In particular, they all feature an informed principal, and hence the strategic transmission of private information and the exercise of control rights are the primary governance mechanisms.

A. Managerial Leadership

Managers and business owners often articulate a strategy that is appropriate given the firm's strategic position and the environment it faces. If managers cannot influence and motivate their subordinates to follow their vision, they may have to exercise their formal authority to bring about change. In this context, communications are effective when the corporate culture creates an environment in which open dialogue can flourish. If communication is effective and valuable, various means of internal communication should be used: in-person meetings, conference calls, emails, internal memos, etc. Applied to this context, the model suggests that managers who micromanage (adopt a hands-off approach) are less (more) likely to be effective communicators. Moreover, since intervention is more difficult in complex and large organizations, the

model predicts that communication as a management tool is more effective in those organizations.

B. Corporate Boards

In a typical public corporation, the CEO runs the company on a daily basis, but the board of directors sets the strategy, approves major decisions, and has the right to replace the CEO and set his compensation. Board members, who are often accomplished and experienced individuals,²⁶ also use their expertise to advise the CEO on a variety of issues (e.g., strategy, public relations, crisis management, and M&A). Board meeting minutes can shed light on the time that directors spend on monitoring, criticizing, and overruling the CEO, as opposed to advising and exchanging of views. Applied to this context, the model predicts a negative association between the monitoring intensity and advisory role of corporate boards. Moreover, since monitoring the CEO requires coordination among directors (e.g., to avoid free-riding), and coordination is more difficult with larger boards that have more diverse and busy directors, the model suggests that these factors would be positively related to the effectiveness of the board's advisory role.

C. Private Equity

PE investors regularly share ideas with their portfolio companies on how to add value. For example, venture capitalists (VCs) advise small start-ups on how to professionalize the management team and commercialize the product,²⁷ and PE shops hire consultants and in-house research teams to help turn around the operations of businesses they acquire.²⁸ At the same time, PE investors often hold board seats and other control/liquidation rights that allow them to intervene in their portfolio companies.²⁹ Interestingly, PE investors routinely invest in multiple firms and syndicate their investments with their peers. In both cases, their ability to intervene is likely diminished: their control rights are diluted, coinvestment creates coordination problems, and diversified portfolios spread them too thinly. Moreover, PE investors are less likely to intervene if exiting their investment is more feasible or profitable, for example, when the IPO and M&A markets are booming. The model predicts that under these scenarios, PE investors would be able to advise and add value to their portfolio companies more effectively.

²⁶ Directors tend to be executives in related industries, such as lawyers, bankers, academics, activist investors (Gow, Shin, and Srinivasan (2014)), or VCs (Celikyurt, Sevilir, and Shivdasani (2014)).

²⁷ See Hellmann and Puri (2000, 2002), Kortum and Lerner (2000), Bottazzi, Rin, and Hellmann (2008), Chemmanur, Krishnan, and Nandy (2011), and Gompers et al. (forthcoming).

²⁸ See Kaplan and Strömberg (2009), Acharya et al. (2013), and Gompers, Kaplan, and Mukharlyamov (2016).

²⁹ See Baker and Gompers (2003), Kaplan and Strömberg (2003, 2004), and Cornelli and Karakas (2015).

D. Shareholder Activism

In a typical campaign, activist investor buys a sizable stake in a public company, expresses her dissatisfaction to the board,³⁰ and explains her view of how the company should be managed.³¹ If the company refuses to comply with the activist's demand, she might launch a public campaign, which could involve a proxy fight to replace the incumbent directors. Applied to this context, the model predicts a negative correlation between activists' interventions and behind-the-scenes communications. Interestingly, factors that facilitate coordination among shareholders (e.g., influential proxy advisers, nondispersed ownership structure) are likely to reduce the cost of campaigning and thereby undermine the ability of activists to influence the policies of their target companies.

IV. Concluding Remarks

Communication and intervention are common remedies for information asymmetries and conflicts of interest in a wide range of applications related to leadership, management, and corporate governance. In this paper, I study a principal-agent model that features both communication and intervention as alternative means of exerting influence. The main result shows that the power of a principal to intervene in an agent's decision exacerbates the underlying agency problem and as a result limits the ability of the principal to use her private information to influence the agent's decision. The perverse effect of intervention on communication is particularly strong when the cost of intervention is low or the underlying agency problem is severe, and in those circumstances intervention is detrimental. Therefore, the power to intervene is least desirable for the principal when intervention is seemingly most effective or needed.

The analysis implies that communication in and of itself can reduce the value of control rights, and as such highlights a novel mechanism through which the allocation of control rights affects real outcomes. As a whole, the analysis of this paper sheds new light on a variety of management and governance applications, and in particular, has implications for the effectiveness of visionary leadership, the tension between the supervisory and advisory roles of corporate boards, and the ability of sophisticated investors to add value to their portfolio companies. Other applications of the model (e.g., to regulation) are left for future research.

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³⁰ For evidence on communications between investors and firms, see Becht et al. (2009), Becht, Franks, and Grant (2015), and McCahery, Sautner, and Starks (2016).

³¹ The idea that insiders can learn from outsiders is central to a new literature that studies how firms use information in stock prices to make investment decisions (e.g., Bond, Edmans, and Goldstein (2012)).

Appendix A: Proofs

SOLUTION CONCEPT: A Perfect Bayesian Equilibrium of the game consists of three parts: the principal's communication strategy $m^*(\theta)$, the agent's decision rule $x^*(m)$, and the principal's intervention policy $\Delta^*(x, \theta)$. The equilibrium is defined as follows: (i) For any realization $\theta \in [\underline{\theta}, \bar{\theta}]$, $m^*(\theta)$ maximizes the expected utility of the principal given the agent's decision rule $x^*(\cdot)$ and the intervention policy $\Delta^*(x, \theta)$; (ii) for any message m , project $x^*(m)$ maximizes the expected utility of the agent, taking into account the principal's communication strategy $m^*(\theta)$ and intervention policy $\Delta^*(x, \theta)$; and (iii) for any realization $\theta \in [\underline{\theta}, \bar{\theta}]$ and $x \in \mathbb{R}$, $\Delta^*(x, \theta)$ maximizes the expected utility of the principal. Finally, all players have rational expectations in that each player's belief about the other players' strategies is correct in equilibrium. Moreover, the agent uses Bayes' Rule to update their beliefs from the principal's message about θ .

LEMMA A.1 (Auxiliary result): *The principal's optimal intervention policy is given by the unique solution of equation (4) and has the following properties:*

- (i) $\Delta^*(0) = 0$ and if $x \neq \theta$, then $|\Delta^*(x - \theta)| < |x - \theta|$.
- (ii) $\Delta^*(x - \theta)$ is a strictly increasing and continuous function of $x - \theta$, where the slope is strictly between zero and one.
- (iii) $|\Delta^*(x_1 - \theta_1) - \Delta^*(x_2 - \theta_2)| < |(x_1 - \theta_1) - (x_2 - \theta_2)|$.
- (iv) $\Delta^*(|x - \theta|) = -\Delta^*(-|x - \theta|)$.
- (v) If $x \neq \theta$, then $|\Delta^*(x - \theta)|$ strictly decreases in δ .

PROOF OF LEMMA A.1: To ease the exposition, let $r \equiv x - \theta$. I first prove part (i). Suppose that $r > 0$ ($r < 0$). The principal never chooses $\Delta > r$ ($\Delta < r$), since by choosing $\Delta = r$ she not only minimizes the $L(\cdot)$ function but also reduces the cost of intervention. Also, the principal never chooses $\Delta < 0$ ($\Delta > 0$), since by choosing $\Delta = 0$ she brings not only the cost of intervention to zero but also reduces the loss from the $L(\cdot)$ function. Therefore, it must be the case that $|\Delta^*| \leq |r|$ and $x = \theta \Rightarrow \Delta^* = 0$, as required by part (i).

Next, I prove that Δ^* is given by the unique solution of equation (4). Suppose $r \leq 0$. Part (i) implies that $r \leq \Delta \leq 0$, and therefore Δ^* solves

$$\max_{r \leq \Delta \leq 0} U_P(\theta, \theta) - L(\Delta - r) - \delta C(-\Delta).$$

The first-order condition implies

$$-L'(\Delta - r) + \delta C'(-\Delta) = 0. \quad (\text{A.1})$$

If $\Delta = 0$, then the left-hand side (LHS) of equation (A.1) is $-L'(-r) + \delta C'(0) = -L'(-r) < 0$. If $\Delta = r$, then the LHS is $-L'(0) + \delta C'(-r) = \delta C'(-r) > 0$. Therefore, from continuity, a solution to equation (A.1) in $(r, 0)$ exists. The second-order condition requires $-L''(\Delta - r) - \delta C''(-\Delta) < 0$. Since $L'' > 0$ and $C'' > 0$, the second-order condition holds and $\Delta^* \in (r, 0)$ is given by the unique solution

of $L'(\Delta - r) = \delta C'(-\Delta)$. Next, suppose $r > 0$. Part (i) implies that $0 \leq \Delta \leq r$, and therefore Δ^* solves

$$\max_{0 \leq \Delta \leq r} U_P(\theta, \theta) - L(r - \Delta) - \delta C(\Delta).$$

The first-order condition implies

$$L'(r - \Delta) - \delta C'(\Delta) = 0. \quad (\text{A.2})$$

If $\Delta = 0$, then the LHS of equation (A.2) is $L'(r) - \delta C'(0) = L'(r) > 0$. If $\Delta = r$, then the LHS is $L'(0) - \delta C'(r) = -\delta C'(r) < 0$. Therefore, from continuity, a solution to equation (A.2) in $(0, r)$ exists. The second-order condition requires $-L''(r - \Delta) - \delta C''(\Delta) < 0$. Therefore, $\Delta^* \in (0, r)$ is given by the unique solution of $L'(r - \Delta) = \delta C'(\Delta)$. Notice that equations (A.1) and (A.2) imply that Δ^* is a continuous function of r . Overall, this proves that Δ^* is the unique solution of equation (4).

Consider part (ii). Applying the implicit function theorem on equation (4), we have

$$\frac{\partial \Delta^*}{\partial r} = \frac{L'(|r - \Delta^*|)}{L'(|r - \Delta^*|) + \delta C''(|\Delta^*|)}. \quad (\text{A.3})$$

Notice that $L'', C'' > 0$ establishes $\frac{\partial \Delta^*}{\partial r} \in (0, 1)$. Since $\frac{\partial \Delta^*}{\partial r} \in (0, 1)$ for all r , it must be the case that $|\Delta^*(r_1) - \Delta^*(r_2)| < |r_1 - r_2|$, which proves part (iii).

Consider part (iv). Suppose without loss of generality that $r > 0$. Recall that $\Delta^*(r)$ is the unique solution of equation (A.2) and $\Delta^*(-r)$ is the unique solution of equation (A.1). Since both equations have a unique solution, it must be the case that $\Delta^*(r) = -\Delta^*(-r)$, as required.

Consider part (v). Applying the implicit function theorem on equation (4), we have

$$\frac{\partial |\Delta^*|}{\partial \delta} = -\frac{C'(|\Delta^*|)}{L'(|r - \Delta^*|) + \delta C''(|\Delta^*|)} < 0, \quad (\text{A.4})$$

as required. ■

LEMMA A.2 (Auxiliary result): $V_P(x, \theta) = U_P(\theta, \theta) - l(|x - \theta|)$, where

$$l(|x - \theta|) \equiv L(|x - \theta - \Delta^*(x - \theta)|) + \delta C(|\Delta^*(x - \theta)|), \quad (\text{A.5})$$

$l'(\cdot) > 0$, and $l(0) = l'(0) = 0$.

PROOF OF LEMMA A.2: According to Lemma A.1, Δ^* depends on x and θ only through $x - \theta$, and so does $l(\cdot)$. To ease the exposition, let $r \equiv x - \theta$. I argue that $l(r) = l(-r)$. According to Lemma A.1, $\Delta^*(r) = -\Delta^*(-r)$ and either $0 < \Delta^*(r) < r$ or $r < \Delta^*(r) < 0$. Therefore, if $r > 0$, it follows that

$$l(r) = L(r - \Delta^*(r)) + \delta C(\Delta^*(r)) = L(\Delta^*(-r) - (-r)) + \delta C(-\Delta^*(-r)) = l(-r),$$

as required. Since $l(r) = l(-r)$, $l(\cdot)$ is effectively a function of $|r|$. Finally, I prove $l'(0) = 0$ and $l'' > 0$. Without loss of generality, suppose $r > 0$. Then

$$\begin{aligned} l'(r) &= L'(r - \Delta^*(r)) \left(1 - \frac{\partial \Delta^*}{\partial r} \right) + \delta C'(\Delta^*(r)) \frac{\partial \Delta^*}{\partial r} \\ &= L'(r - \Delta^*(r)). \end{aligned}$$

The second equality follows from the identity $L'(r - \Delta^*(r)) = \delta C'(\Delta^*(r))$. Since $\Delta^*(0) = 0$, $l'(0) = L'(0) = 0$. Notice that $l''(r) = L''(r - \Delta^*(r))(1 - \frac{\partial \Delta^*}{\partial r})$. Since $L'' > 0$ and $\frac{\partial \Delta^*}{\partial r} \in (0, 1)$, it follows that $l'' > 0$, as required. ■

PROOF OF LEMMA 1: I prove $\frac{\partial V_A(x, \theta; b)}{\partial x} > 0 \Leftrightarrow x < \theta + \beta$. To ease the exposition, let $r \equiv x - \theta$. I proceed in several steps:

(1) Suppose that $r \leq 0$. According to Lemma A.1, $r \leq \Delta^*(r) \leq 0$. Therefore, $V_A(x, \theta; b) = U_A(\theta + b, \theta) - T(\Delta^*(r) - r + b)$ and

$$\frac{\partial V_A(x, \theta; b)}{\partial x} = T'(\Delta^*(r) - r + b) \left(1 - \frac{\partial \Delta^*}{\partial r} \right).$$

Since $\frac{\partial \Delta^*}{\partial r} \in (0, 1)$, $\Delta^*(r) - r + b > 0$, $T'(0) = 0$, and $T'' > 0$, it must be $\frac{\partial V_A(x, \theta; b)}{\partial x} > 0$.

(2) Suppose that $r > 0$ and $b < r - \Delta^*(r)$. According to Lemma A.1, $0 < \Delta^*(r) < r$. Since $b < r - \Delta^*(r)$, in this region $V_A(x, \theta; b) = U_A(\theta + b, \theta) - T(r - b - \Delta^*(r))$ and

$$\frac{\partial V_A(x, \theta; b)}{\partial x} = -T'(r - b - \Delta^*(r)) \left(1 - \frac{\partial \Delta^*}{\partial r} \right).$$

Since $\frac{\partial \Delta^*}{\partial r} \in (0, 1)$, $b < r - \Delta^*(r)$, $T'(0) = 0$, and $T'' > 0$, it must be the case that $\frac{\partial V_A(x, \theta; b)}{\partial x} < 0$.

(3) Suppose that $r > 0$ and $r - \Delta^*(r) \leq b$. According to Lemma A.1, $0 < \Delta^*(r) < r$. Therefore, in this region $V_A(x, \theta; b) = U_A(\theta + b, \theta) - T(b + \Delta^*(r) - r)$ and

$$\frac{\partial V_A(x, \theta; b)}{\partial x} = T'(b + \Delta^*(r) - r) \left(1 - \frac{\partial \Delta^*}{\partial r} \right).$$

Since $T'(0) = 0$ and $T'' > 0$, if $b + \Delta^*(r) - r > 0$, we have $\frac{\partial V_A(x, \theta; b)}{\partial x} > 0$, while if $b + \Delta^*(r) - r = 0$, we have $\frac{\partial V_A(x, \theta; b)}{\partial x} = 0$.

(4) Based on the three steps above, if there exists a unique $r^* > 0$ such that $r^* - \Delta^*(r^*) = b$, then it is the unique maximum of $V_A(x, \theta; b)$. Otherwise, $\frac{\partial V_A(x, \theta; b)}{\partial x} > 0$ for all r . I argue that

$$r^* - \Delta^*(r^*) = b$$

has a unique solution that is given by β . Recall that $\Delta^*(0) = 0$. Then, $0 - \Delta^*(0) = 0$. Further recall that $\frac{\partial \Delta^*(r)}{\partial r} \in (0, 1)$. Then $r - \Delta^*(r)$ is a strictly increasing function. Next, consider $\lim_{r \rightarrow \infty} [r - \Delta^*(r)]$. Note that according to Lemma A.1, if

$r > 0$, it follows that $\Delta^*(r)$ is the unique solution of $L'(r - \Delta) = \delta C'(\Delta)$. Therefore, we can represent $\Delta^*(r)$ as $r = \Delta^*(r) + (L')^{-1}(\delta C'(\Delta^*(r)))$. Since $(L')^{-1}(\delta C'(\cdot))$ is a strictly increasing function, it must be the case that $\lim_{r \rightarrow \infty} \Delta^*(r) = \infty$, which implies that $\lim_{r \rightarrow \infty} [r - \Delta^*(r)] = \lim_{r \rightarrow \infty} (L')^{-1}(\delta C'(\Delta^*(r)))$. Therefore, if $\lim_{r \rightarrow \infty} (L')^{-1}(\delta C'(\Delta^*(r))) > b$, it follows that $r - \Delta^*(r) = b$ has a unique solution and is given by β . If true, the unique maximum of $V_A(x, \theta; b)$ is obtained at $x = \theta + \beta$. I prove that if $\lim_{r \rightarrow \infty} C''(r) > 0$, then $\lim_{r \rightarrow \infty} (L')^{-1}(\delta C'(\Delta^*(r))) = \infty$. To see why, suppose on the contrary that the limit is finite. Then it must be the case that either $\lim_{r \rightarrow \infty} (L')^{-1}(r) < \infty$ or $\lim_{r \rightarrow \infty} C'(r) < \infty$. Notice that if $\lim_{x \rightarrow \infty} f(x)$ is finite and $\lim_{x \rightarrow \infty} f'(x)$ exists, then $\lim_{x \rightarrow \infty} f'(x) = 0$. Therefore, $\lim_{r \rightarrow \infty} (L')^{-1}(r) < \infty$ requires that

$$\lim_{r \rightarrow \infty} [(L')^{-1}(r)]' = 0 \Leftrightarrow \lim_{r \rightarrow \infty} \frac{1}{L''((L')^{-1}(r))} = 0.$$

However, $\lim_{r \rightarrow \infty} (L')^{-1}(r) < \infty$ and $L'' > 0$ implies that $\lim_{r \rightarrow \infty} \frac{1}{L''((L')^{-1}(r))} > 0$, a contradiction. Therefore, it must be the case that $\lim_{r \rightarrow \infty} C'(r) < \infty$. However, the assumption $\lim_{r \rightarrow \infty} C''(r) > 0$ guarantees that it cannot be the case that $\lim_{r \rightarrow \infty} C'(r) < \infty$, also a contradiction. Thus, we must have $\lim_{r \rightarrow \infty} (L')^{-1}(\delta C'(\Delta^*(r))) = \infty$, as required.³² ■

PROOF OF THEOREM 1: The proof has several steps.

- (1) Recall Lemma A.2 proves that $V_P(x, \theta)$ obtains its unique maximum at θ and Lemma 1 proves that $V_A(x, \theta; b)$ obtains its unique maximum at $\theta + \beta$. Also, recall the focus on pure strategies (or alternatively, the additional assumption that $V_A(x, \theta; b)$ is concave in x). Therefore, a variant of Theorem 1 in Crawford and Sobel (1982) shows that in a communication phase (in a game without intervention), there exists a positive and finite integer N^* such that for every $n \in \{1, \dots, N^*\}$ there exists at least one equilibrium, where

$$\bar{x}(\underline{a}, \bar{a}) \equiv \arg \max_x E[V_A(x, \theta; b) | \theta \in (\underline{a}, \bar{a})], \quad (\text{A.6})$$

$$\mu^*(\bar{x}(a_{i-1}, a_i) | \theta) = 1 \quad \text{if } \theta \in (a_{i-1}, a_i), \quad (\text{A.7})$$

$$x^*(m) = \bar{x}(a_{i-1}, a_i) \quad \text{if } m \in (a_{i-1}, a_i), \quad (\text{A.8})$$

$$V_P(\bar{x}(a_i, a_{i+1}), a_i) - V_P(\bar{x}(a_{i-1}, a_i), a_i) = 0; \quad \text{for } i = 1, \dots, n-1, \quad (\text{A.9})$$

$$a_0 = \underline{\theta} \quad \text{and} \quad a_n = \bar{\theta}. \quad (\text{A.10})$$

³² If $\lim_{r \rightarrow \infty} C''(r) = 0$ and b is sufficiently large, then $\frac{\partial}{\partial x} V_A(x, \theta; b) > 0$ for all x , that is, the agent will choose the largest project possible irrespective of his beliefs about θ . Therefore, in equilibrium, all messages must result in the same action. In some sense, the agent behaves as if $\beta = \infty$. Clearly, in those cases the equilibrium is uninformative, and hence intervention hinders communication.

Moreover, all other equilibria are economically equivalent to those in this class for some value of $n \in \{1, \dots, N^*\}$. In other words, the set of equilibria and their properties are determined by the functional form of condition (A.6) and the solution of equation (A.9).

- (2) I argue that if $\theta_0 \in [\underline{\theta}, \bar{\theta}]$ and $x_1 < x_2$, then $V_P(x_1, \theta_0) = V_P(x_2, \theta_0)$ implies that $\theta_0 = \frac{x_1 + x_2}{2}$. Indeed, according to Lemma A.2, $V_P(x, \theta) = U_P(\theta, \theta) - l(|x - \theta|)$, where $l'(0) = 0$ and $|x - \theta| > 0 \Rightarrow l'(|x - \theta|) > 0$. Therefore, $V_P(x_1, \theta_0) = V_P(x_2, \theta_0)$ implies that $l(|x_1 - \theta_0|) = l(|x_2 - \theta_0|)$. Since $l(\cdot)$ is a strictly increasing function, it must be the case that $|x_1 - \theta_0| = |x_2 - \theta_0|$, which implies $\theta_0 = \frac{x_1 + x_2}{2}$, as required.
- (3) I argue that $\bar{x}(\underline{a}, \bar{a})$ is the unique solution of

$$x - \underline{a} - \Delta^*(x - \underline{a}) - b = -(x - \bar{a} - \Delta^*(x - \bar{a}) - b). \quad (\text{A.11})$$

Since θ is uniformly distributed, condition (A.6) implies that

$$\bar{x}(\underline{a}, \bar{a}) \equiv \arg \min_x \frac{1}{\bar{a} - \underline{a}} \int_{\underline{a}}^{\bar{a}} T(|x - \theta - \Delta^*(x - \theta) - b|) d\theta.$$

Recall from the proof of Lemma 1 that $x - \theta - \Delta^*(x - \theta) - b > 0$ if and only if $x - \theta > \beta$. Therefore,

$$\int_{\underline{a}}^{\bar{a}} T(|x - \theta - \Delta^*(x - \theta) - b|) d\theta = \left[\int_{\min\{\underline{a}, x - \beta\}}^{\min\{\bar{a}, x - \beta\}} T(x - \theta - \Delta^*(x - \theta) - b) d\theta + \int_{\max\{\underline{a}, x - \beta\}}^{\max\{\bar{a}, x - \beta\}} T(-(x - \theta - \Delta^*(x - \theta) - b)) d\theta \right].$$

Notice that if $x - \beta \in [\underline{a}, \bar{a}]$, it follows that the derivative of the integral boundaries with respect to x are zero, as

$$T(|x - (x - \beta) - \Delta^*(x - (x - \beta)) - b|) = T(|\beta - \Delta^*(\beta) - b|) = T(0) = 0.$$

Therefore,

$$\begin{aligned} & \frac{\partial}{\partial x} \int_{\underline{a}}^{\bar{a}} T(|x - \theta - \Delta^*(x - \theta) - b|) d\theta \\ &= \left[\int_{\min\{\underline{a}, x - \beta\}}^{\min\{\bar{a}, x - \beta\}} T'(x - \theta - \Delta^*(x - \theta) - b) \left(1 - \frac{\partial \Delta^*(x - \theta)}{\partial x}\right) d\theta + \int_{\max\{\underline{a}, x - \beta\}}^{\max\{\bar{a}, x - \beta\}} T'(-(x - \theta - \Delta^*(x - \theta) - b)) \left(\frac{\partial \Delta^*(x - \theta)}{\partial x} - 1\right) d\theta \right]. \end{aligned}$$

Since $\frac{\partial \Delta^*(x - \theta)}{\partial x} \in (0, 1)$ and $T' > 0$, the derivative can be zero only if $x - \beta \in [\underline{a}, \bar{a}]$. Therefore, it must be the case that $\bar{x}(\underline{a}, \bar{a}) \in [\underline{a} + \beta, \bar{a} + \beta]$. Integrating the derivative with respect to θ , we get

$$\begin{aligned} & \frac{\partial}{\partial x} \int_{\underline{a}}^{\bar{a}} T(|x - \theta - \Delta^*(x - \theta) - b|) d\theta \\ &= \left[-\left(T(\beta - \Delta^*(\beta) - b) - T(x - \underline{a} - \Delta^*(x - \underline{a}) - b)\right) - \left(T(-(x - \bar{a} - \Delta^*(x - \bar{a}) - b)) - T(-(\beta - \Delta^*(\beta) - b))\right) \right] \\ &= T(x - \underline{a} - \Delta^*(x - \underline{a}) - b) - T(-(x - \bar{a} - \Delta^*(x - \bar{a}) - b)). \end{aligned}$$

Since $\frac{\partial \Delta^*(x-\theta)}{\partial x} \in (0, 1)$, we have that $x - \underline{a} - \Delta^*(x - \underline{a}) - b$ increases in x and $-(x - \bar{a} - \Delta^*(x - \bar{a}) - b)$ decreases in x . Moreover, since $T' > 0$, we have that $\frac{\partial}{\partial x} \int_{\underline{a}}^{\bar{a}} T(|x - \theta - \Delta^*(x - \theta) - b|) d\theta$ increases in x . Therefore, if $\frac{\partial}{\partial x} \int_{\underline{a}}^{\bar{a}} T(|x - \theta - \Delta^*(x - \theta) - b|) d\theta = 0$ has a solution, the solution must be unique and it must solve equation (A.11). Moreover, $\bar{x}(\underline{a}, \bar{a})$ must be this solution. It is left to show that the solution equation (A.11) does indeed exist. If $x = \underline{a} + \beta$, then the LHS of equation (A.11) is zero and the right-hand side (RHS) of equation (A.11) is positive. Similarly, if $x = \bar{a} + \beta$, then the RHS of equation (A.11) is zero and the LHS of equation (A.11) is positive. Therefore, a unique solution exists, as required. Moreover, $\bar{x}(\underline{a}, \bar{a})$ must be the unique solution of equation (A.11), the global maximizer of $E[V_A(x, \theta; b) | \theta \in (\underline{a}, \bar{a})]$.

- (4) Fourth, I argue that $\bar{x}(\underline{a}, \bar{a}) > \frac{\bar{a} + \underline{a}}{2} + b$. Since the LHS of equation (A.11) is increasing in x and the RHS is decreasing, it is sufficient to show that the LHS of equation (A.11) is smaller than the RHS when $x = \frac{\bar{a} + \underline{a}}{2} + b$. Indeed, if $x = \frac{\bar{a} + \underline{a}}{2} + b$, then the LHS of equation (A.11) is $\frac{\bar{a} - \underline{a}}{2} - \Delta^*(\frac{\bar{a} - \underline{a}}{2} + b)$ and the RHS is $\frac{\bar{a} - \underline{a}}{2} + \Delta^*(b - \frac{\bar{a} - \underline{a}}{2})$. Thus, the LHS of equation (A.11) is smaller than the RHS if and only if $-\Delta^*(b + \frac{\bar{a} - \underline{a}}{2}) < \Delta^*(b - \frac{\bar{a} - \underline{a}}{2})$. Since $\Delta^*(r) = -\Delta^*(-r)$, the previous inequality holds if and only if $\Delta^*(-b - \frac{\bar{a} - \underline{a}}{2}) < \Delta^*(b - \frac{\bar{a} - \underline{a}}{2})$. Since $\Delta^*(\cdot)$ is a strictly increasing function and $b > 0$, the LHS of equation (A.11) is smaller than the RHS. Therefore, $\bar{x}(\underline{a}, \bar{a}) > \frac{\bar{a} + \underline{a}}{2} + b$, as required.
- (5) I argue that the partition in equilibrium must satisfy $a_{i+1} - a_i < a_i - a_{i-1} - 4b$. Together with the second step, equation (A.9) implies that $a_i = \frac{\bar{x}(a_{i-1}, a_i) + \bar{x}(a_i, a_{i+1})}{2}$ for $i = 1, \dots, n-1$. Together with the fourth step, which proves $\bar{x}(a_{i-1}, a_i) > \frac{a_{i-1} + a_i}{2} + b$, it must be the case that $a_{i+1} - a_i < a_i - a_{i-1} - 4b$.
- (6) I prove $N_{in}^* \leq N_{no}^*$. Consider the game without intervention. The five steps above can be repeated with the exception that no intervention is equivalent to assuming that $\Delta^*(x - \theta) \equiv 0$. The first and second steps continue to hold. The third and fourth steps imply that without intervention, $\bar{x}(\underline{a}, \bar{a}) = \frac{\bar{a} + \underline{a}}{2} + b$. The fifth step implies that $a_{i+1} - a_i = a_i - a_{i-1} - 4b$, that is, condition (10) holds with equality. Similar to the solution of the leading example of Crawford and Sobel (1982), the solution to this indifference equation is parameterized by a_{N-1} (given the boundary condition $a_N = \bar{\theta}$) and is given by

$$a_i = \bar{\theta} - (N - i) \cdot (\bar{\theta} - a_{N-1}) - 2(N - i)(N - 1 - i)b.$$

The largest number of intervals that can be supported in equilibrium (without intervention) is the largest integer N such that $a_{N-1} = \bar{\theta}$ and $a_0 > \underline{\theta}$, that is, the largest N such that $\bar{\theta} - 2N(N - 1)b > \underline{\theta}$. Therefore, $N_{no}^* = N(b)$, where the function $N(\cdot)$ is defined in equation (12). Repeating

similar arguments as in the communication game with intervention, the inequality condition (10) implies that for a given a_{N-1} ,

$$a_i < \bar{\theta} - (N-i) \cdot (\bar{\theta} - a_{N-1}) - 2(N-i)(N-1-i)b. \quad (A.12)$$

Therefore, $a_0 < \bar{\theta} - N \cdot (\bar{\theta} - a_{N-1}) - 2N(N-1)b$. Requiring $a_0 \geq \underline{\theta}$ implies that

$$N \cdot (\bar{\theta} - a_{N-1}) + 2N(N-1)b < \bar{\theta} - \underline{\theta}.$$

Therefore, the largest number of intervals that can be supported in equilibrium (with intervention) has to satisfy $2N(N-1)b < \bar{\theta} - \underline{\theta}$. Since N_{no}^* is the *largest* integer that satisfies $2N(N-1)b < \bar{\theta} - \underline{\theta}$, we have $N_{in}^* \leq N_{no}^*$, as required.

- (7) According to the sixth step, $N_{in}^* \leq N_{no}^* = N(b)$. If $N(b) = 1$, then $N_{in}^* = 1$, and $\Sigma_{in}^* = \Sigma_{no}^* = \sigma^2$, that is, no information is revealed in both cases. Suppose that $N(b) \geq 2$. Crawford and Sobel (1982) prove that if $a_{i+1} - a_i = a_i - a_{i-1} - 4b$, θ is uniformly distributed, and the partition has n elements, then $\Sigma = \Sigma(n)$, where

$$\Sigma(n) \equiv \frac{\sigma^2}{n^2} + \frac{b^2(n^2 - 1)}{3}.$$

Notice that $\Sigma(n)$ is a decreasing function of n if and only if $b < \frac{\bar{\theta} - \underline{\theta}}{2n}$. Also note that $N(b) \geq 2$ if and only if $b < \frac{\bar{\theta} - \underline{\theta}}{4}$. Therefore, if $N(b) \geq 2$, it follows that $\Sigma(n)$ is a decreasing function of n , and as a result, $\Sigma_{no}^* = \Sigma(N(b))$. Moreover, since $N_{in}^* \leq N(b)$, we have $\Sigma(N_{in}^*) \geq \Sigma_{no}^*$. I prove $\Sigma_{in}^* > \Sigma(N_{in}^*)$, which will complete the argument. Note that given partition $(a_0, \dots, a_{n-1}, a_N)$,

$$\begin{aligned} E \left[(E[\theta | m^*(\theta)] - \theta)^2 \right] &= \sum_{i=1}^N \Pr[\theta \in (a_{i-1}, a_i)] \mathbb{E}_i \left[(\mathbb{E}_i[\theta] - \theta)^2 \right] \\ &= \sum_{i=1}^N \frac{a_i - a_{i-1}}{\bar{\theta} - \underline{\theta}} \left(\frac{1}{a_i - a_{i-1}} \int_{a_{i-1}}^{a_i} \left[\frac{a_{i-1} + a_i}{2} - \theta \right]^2 d\theta \right) \\ &= \sum_{i=1}^N \frac{1}{\bar{\theta} - \underline{\theta}} \left(-\frac{1}{3} \left[\frac{a_{i-1} + a_i}{2} - a_i \right]^3 + \frac{1}{3} \left[\frac{a_{i-1} + a_i}{2} - a_{i-1} \right]^3 \right) \\ &= \frac{1}{\bar{\theta} - \underline{\theta}} \frac{1}{12} \sum_{i=1}^N (a_i - a_{i-1})^3. \end{aligned}$$

³³ This claim can be proved by induction using inequality (10). It is easy to see that inequality (10) implies that (A.12) holds for $i = N - 2$. Suppose that (A.12) holds for $i \geq k$. Then (A.12) also holds for $i - 1$. Indeed, note that (10) implies that $a_N - a_{N-1} + (N-i)4b - a_i < -a_{i-1}$. Applying (A.12) to $-a_i$ shows that (A.12) also holds for a_{i-1} .

Let $(a_0, \dots, a_{N-1}, a_N)$ and $(a'_0, \dots, a'_{N-1}, a'_N)$ be a partition of size N in equilibrium without and with intervention, respectively. Then, $\Sigma_{in}^* = \min_{N \in \{1, \dots, N_{in}^*\}} \sum_{i=1}^N (a_i - a_{i-1})^3$ and $\Sigma_{no}^* = \frac{1}{\theta - \underline{\theta}} \frac{1}{12} \sum_{i=1}^{N(b)} (a_i - a_{i-1})^3$. I prove that for every $N \in \{1, \dots, N_{in}^*\}$, we have $\sum_{i=1}^N (a'_i - a'_{i-1})^3 > \sum_{i=1}^N (a_i - a_{i-1})^3$, which implies $\Sigma_{in}^* > \Sigma_{no}^*$.

Fix $N \in \{1, \dots, N_{in}^*\}$. There exists $i^* \in \{1, \dots, N-1\}$ such that if $i \leq i^*$, then $(a'_i - a'_{i-1}) > (a_i - a_{i-1})$, whereas if $i > i^*$, then $(a'_i - a'_{i-1}) < (a_i - a_{i-1})$. To see why, first note that both partitions have N elements and both span the interval $[\underline{\theta}, \bar{\theta}]$. The claim follows from the observation that without intervention, the size of the intervals decreases by $4b$, while with intervention, it decreases by a rate strictly larger than $4b$. Next, for every $i \in \{1, \dots, i^*\}$, define $p_i = (a'_i - a'_{i-1}) - (a_i - a_{i-1})$, and for every $k \in \{i^* + 1, \dots, N\}$, define $q_k = (a_k - a_{k-1}) - (a'_k - a'_{k-1})$. Consider the following algorithm that deforms $(a'_0, \dots, a'_{N-1}, a'_N)$ to $(a_0, \dots, a_{N-1}, a_N)$:

Set $i = 1$ and $k = i^* + 1$,

If $i \leq i^*$ or $k \leq N$, do exactly one of the following:

- (a) If $p_i \leq q_k$, set $(a'_i - a'_{i-1}) \mapsto (a'_i - a'_{i-1}) - p_i$, $(a'_k - a'_{k-1}) \mapsto (a'_k - a'_{k-1}) + p_i$, $q_k \mapsto q_k - p_i$, and $i \mapsto i + 1$.
- (b) If $p_i > q_k$, set $(a'_k - a'_{k-1}) \mapsto (a'_k - a'_{k-1}) + q_k$, $(a'_i - a'_{i-1}) \mapsto (a'_i - a'_{i-1}) - q_k$, $p_i \mapsto p_i - q_k$, and $k \mapsto k + 1$.

In step (a), $(a'_i - a'_{i-1})$ decreases by p_i to $(a_i - a_{i-1})$, and $(a'_k - a'_{k-1})$ increases by p_i to a term that is still smaller than $(a_k - a_{k-1})$. The term $q_k - p_i$ records the difference between $(a_k - a_{k-1})$ and the adjusted $(a'_k - a'_{k-1})$. Since i increases by one, the interval $(a'_i - a'_{i-1})$ will no longer be treated by the algorithm. Since k does not change, the adjusted $(a'_k - a'_{k-1})$, which is still smaller than $(a_k - a_{k-1})$, will be treated in the next iteration. Note that due to this deformation, it follows that

$$\begin{aligned} (a'_i - a'_{i-1})^3 + (a'_k - a'_{k-1})^3 &> (a'_i - a'_{i-1} - p_i)^3 + (a'_k - a'_{k-1} + p_i)^3 \Leftrightarrow \\ (a'_i - a'_{i-1}) - (a'_k - a'_{k-1}) &> p_i \Leftrightarrow \\ (a'_i - a'_{i-1}) - (a'_k - a'_{k-1}) &> (a'_i - a'_{i-1}) - (a_i - a_{i-1}) \Leftrightarrow \\ (a_i - a_{i-1}) &> (a'_k - a'_{k-1}). \end{aligned}$$

But notice that $i < k$ implies that $(a_i - a_{i-1}) > (a_k - a_{k-1})$,³⁴ and $i^* + 1 \leq k$ implies that $(a_k - a_{k-1}) > (a'_k - a'_{k-1})$. Therefore, $(a_i - a_{i-1}) > (a'_k - a'_{k-1})$, that is, $\sum_{i=1}^N (a'_i - a'_{i-1})^3$ decreases due to this step of the deformation.

In step (b), $(a'_k - a'_{k-1})$ increases by q_k to $(a_k - a_{k-1})$, and $(a'_i - a'_{i-1})$ decreases by q_k , which is still greater than $(a_i - a_{i-1})$. The term $p_i - q_k$ records the difference between the adjusted $(a'_i - a'_{i-1})$ and $(a_i - a_{i-1})$. Since k increases by one,

³⁴ Recall the size of the intervals in partition of the game without intervention is decreasing by $4b$. Note that this partition is not treated by the algorithm.

the interval $(a'_k - a'_{k-1})$ will no longer be treated by the algorithm. Since i does not change, the adjusted $(a'_i - a'_{i-1})$, which is still larger than $(a_i - a_{i-1})$, will be treated in the next iteration. Note that due to this deformation, we have

$$\begin{aligned} (a'_i - a'_{i-1})^3 + (a'_k - a'_{k-1})^3 &> (a'_i - a'_{i-1} - q_i)^3 + ((a'_k - a'_{k-1}) + q_i)^3 \Leftrightarrow \\ (a'_i - a'_{i-1}) - (a'_k - a'_{k-1}) &> q_i \Leftrightarrow \\ (a'_i - a'_{i-1}) - (a'_k - a'_{k-1}) &> (a_k - a_{k-1}) - (a'_k - a'_{k-1}) \Leftrightarrow \\ (a'_i - a'_{i-1}) &> (a_k - a_{k-1}). \end{aligned}$$

But notice that $i \leq i^*$ implies that $(a'_i - a'_{i-1}) > (a_i - a_{i-1})$ and $i < k$ implies that $(a_i - a_{i-1}) > (a_k - a_{k-1})$. Therefore, $(a'_i - a'_{i-1}) > (a_k - a_{k-1})$, that is, $\sum_{i=1}^N (a'_i - a'_{i-1})^3$ decreases due to this step of the deformation.

Finally, notice that in each iteration of the algorithm either i increases by one or k increases by one, but never both. Therefore, the algorithm stops after exactly N iterations. Moreover, in each of these steps, a distinct interval of the deformed partition is treated by the algorithm and becomes identical to the corresponding interval in the partition $(a_0, \dots, a_{N-1}, a_N)$. Therefore, when the algorithm ends, $(a'_i - a'_{i-1}) = (a_i - a_{i-1})$ for all $i \in \{1, \dots, N\}$. Since $\sum_{i=1}^N (a'_i - a'_{i-1})^3$ decreases in each step, $\sum_{i=1}^N (a'_i - a'_{i-1})^3 > \sum_{i=1}^N (a_i - a_{i-1})^3$, as required. ■

PROOF OF PROPOSITION 1: Consider first the result in part (i) with respect to δ . The proof has five steps:

- (1) I argue that $\lim_{\delta \rightarrow 0} \beta = \infty$. Indeed, recall that β satisfies $L(b) = \delta C'(\Delta^*(\beta))$. Since $L(b) > 0$, as $\delta \rightarrow 0$, it must be the case that $C'(\Delta^*(\beta)) \rightarrow \infty$, which implies that $\Delta^*(\beta) \rightarrow \infty$. Since $\lim_{\delta \rightarrow 0} \Delta^*(\beta) = \infty$ and β also satisfies $\beta = b + \Delta^*(\beta)$, we have $\lim_{\delta \rightarrow 0} \beta = \infty$.
- (2) Based on step (1), $\lim_{\delta \rightarrow 0} \Delta^*(\beta) = \infty$. Since $\beta = b + \Delta^*(\beta)$ and $\Delta^*(\beta)$ is independent of $\bar{\theta}$ and $\underline{\theta}$, without loss of generality I focus on δ sufficiently small such that $\underline{\theta} + \beta > \bar{\theta} + b$. Recall from the proof of Theorem 1 that if the agent believes that $\theta \in [\underline{a}, \bar{a}]$, then he chooses $x > \underline{a} + \beta$. Since $\underline{a} \geq \underline{\theta}$, $\underline{\theta} + \beta > \bar{\theta}$ implies that regardless of the agent's beliefs about θ , his optimal choice is above $\bar{\theta}$. Therefore, the principal always has strict incentives to induce the smallest project on the equilibrium path, which implies that the equilibrium is not informative, that is, $N_{in}^* = 1$.
- (3) Since U_P^{**} is independent of δ , then if $\lim_{\delta \rightarrow 0} V_P^{**} = -\infty$, there exists $\bar{\delta} > 0$ such that if $\delta < \bar{\delta}$, then $N_{in}^* = 1$ and $V_P^{**} < U_P^{**}$. Since $x^* > \bar{\theta}$, we have $x^* - \theta > 0$ for all θ . Based on Lemma A.1, $x^* - \theta > 0$ implies that $\Delta^*(x^* - \theta) > 0$ and $x^* - \theta - \Delta^*(x^* - \theta) > 0$. Therefore,

$$V_P^{**} = E[U_P(\theta, \theta)] - E[L(x^* - \theta - \Delta^*(x^* - \theta))] - \delta E[C(\Delta^*(x^* - \theta))].$$

Since $x^* > \underline{\theta} + \beta$ and $\Delta^*(\cdot)$ is an increasing function,

$$V_P^{**} \leq E[U_P(\theta, \theta)] - E[L(x^* - \theta - \Delta^*(x^* - \theta))] - \delta C(\Delta^*(\underline{\theta} + \beta - \theta)).$$

Since $-E[L(x^* - \theta - \Delta^*(x^* - \theta))] \leq 0$, proving $\lim_{\delta \rightarrow 0} \delta C(\Delta^*(\beta - (\bar{\theta} - \underline{\theta}))) = \infty$ proves $\lim_{\delta \rightarrow 0} V_P^{**} = -\infty$.

- (4) I prove that if $\lim_{\delta \rightarrow 0} \delta C(\Delta^*(\beta)) = \infty$, we have $\lim_{\delta \rightarrow 0} \delta C(\Delta^*(\beta - (\bar{\theta} - \underline{\theta}))) = \infty$. To see why, recall that according to Lemma A.1, $\Delta^*(\beta) - \Delta^*(\beta - (\bar{\theta} - \underline{\theta})) \in (0, \bar{\theta} - \underline{\theta})$. Therefore,

$$\lim_{\delta \rightarrow 0} \frac{\delta C(\Delta^*(\beta - (\bar{\theta} - \underline{\theta})))}{\delta C(\Delta^*(\beta))} = \lim_{\delta \rightarrow 0} \frac{C(\Delta^*(\beta - (\bar{\theta} - \underline{\theta})))}{C(\Delta^*(\beta))} \geq \lim_{\delta \rightarrow 0} \frac{C(\Delta^*(\beta) - (\bar{\theta} - \underline{\theta}))}{C(\Delta^*(\beta))}.$$

Recall that $\lim_{\delta \rightarrow 0} \Delta^*(\beta) = \infty$. Since $C', C'' > 0$, we have $\lim_{\delta \rightarrow 0} C(\Delta^*(\beta)) = \infty$ and $\lim_{\delta \rightarrow 0} C(\Delta^*(\beta) - (\bar{\theta} - \underline{\theta})) = \infty$. From L'Hospital's Rule,

$$\lim_{\delta \rightarrow 0} \frac{C(\Delta^*(\beta) - (\bar{\theta} - \underline{\theta}))}{C(\Delta^*(\beta))} = \lim_{\delta \rightarrow 0} \frac{C'(\Delta^*(\beta) - (\bar{\theta} - \underline{\theta}))}{C'(\Delta^*(\beta))}.$$

Since there is a finite $\kappa \geq 2$ such that $\lim_{\Delta \rightarrow \infty} \frac{\partial^\kappa C(\Delta)}{\partial^\kappa \Delta} \in (-\infty, +\infty)$, it must be the case that

$$\frac{\lim_{\delta \rightarrow 0} \frac{\partial^\kappa C'(\Delta^*(\beta) - (\bar{\theta} - \underline{\theta}))}{\partial^\kappa \Delta}}{\lim_{\delta \rightarrow 0} \frac{\partial^\kappa C'(\Delta^*(\beta))}{\partial^\kappa \Delta}} = 1.$$

Therefore, $\lim_{\delta \rightarrow 0} \frac{\delta C(\Delta^*(\beta - (\bar{\theta} - \underline{\theta})))}{\delta C(\Delta^*(\beta))} \geq 1$, which implies that if $\lim_{\delta \rightarrow 0} \delta C(\Delta^*(\beta)) = \infty$, then $\lim_{\delta \rightarrow 0} \delta C(\Delta^*(\beta - (\bar{\theta} - \underline{\theta}))) = \infty$.

- (5) I prove $\lim_{\delta \rightarrow 0} \delta C(\Delta^*(\beta)) = \infty$. Recall that $\Delta^*(\beta) = (C')^{-1}(\frac{L'(b)}{\delta})$. Since $C', C'' > 0$ and $\lim_{\delta \rightarrow 0} \Delta^*(\beta) = \infty$, L'Hospital's Rule implies that

$$\begin{aligned} \lim_{\delta \rightarrow 0} \delta C(\Delta^*(\beta)) &= \lim_{\delta \rightarrow 0} \frac{C'(\Delta^*(\beta))}{-\frac{1}{\delta^2}} \times \frac{\partial \Delta^*(\beta)}{\partial \delta} \\ &= \lim_{\delta \rightarrow 0} \frac{C'((C')^{-1}(\frac{L'(b)}{\delta}))}{-\frac{1}{\delta^2}} \times \frac{\partial (C')^{-1}(\frac{L'(b)}{\delta})}{\partial \delta} \\ &= \lim_{\delta \rightarrow 0} \frac{L'(b)}{-\frac{1}{\delta}} \times \frac{\frac{-L'(b)}{\delta^2}}{C''((C')^{-1}(\frac{L'(b)}{\delta}))} = \lim_{\delta \rightarrow 0} \frac{(L'(b))^2}{\delta C''(\Delta^*(\beta))}. \end{aligned}$$

If $\lim_{r \rightarrow \infty} C''(r) < \infty$, then $\lim_{\delta \rightarrow 0} \frac{(L'(b))^2}{\delta C''(\Delta^*(\beta))} = \infty$, as required. Suppose that $\lim_{r \rightarrow \infty} C''(r) = \infty$. From L'Hospital's Rule,

$$\lim_{\delta \rightarrow 0} \frac{\frac{1}{\delta}}{C''(\Delta^*(\beta))} = \lim_{\delta \rightarrow 0} \frac{-\frac{1}{\delta^2}}{C'''(\Delta^*(\beta)) \frac{-L'(b)}{\delta^2 C''(\Delta^*(\beta))}} = \lim_{\delta \rightarrow 0} \frac{C''(\Delta^*(\beta))}{C'''(\Delta^*(\beta))} \frac{1}{L'(b)}.$$

If $\lim_{r \rightarrow \infty} C'''(r) < \infty$, then $\lim_{\delta \rightarrow 0} \frac{C''(\Delta^*(\beta))}{C'''(\Delta^*(\beta))} = \infty$, as required. Suppose that $\lim_{r \rightarrow \infty} C'''(r) = \infty$. From L'Hospital's Rule,

$$\lim_{\delta \rightarrow 0} \frac{C''(\Delta^*(\beta))}{C'''(\Delta^*(\beta))} = \lim_{\delta \rightarrow 0} \frac{C'''(\Delta^*(\beta))}{C''''(\Delta^*(\beta))}.$$

As long as there exists a finite κ such that $\lim_{r \rightarrow \infty} \frac{\partial^\kappa C(r)}{\partial^\kappa r} < \infty$, the same argument can be repeated κ times to conclude that $\lim_{\delta \rightarrow 0} \delta C(\Delta^*(\beta)) = \infty$, which completes the proof.

Next, consider the result in part (i) with respect to b . The proof has seven steps:

- (1) Since $\beta = b + \Delta^*(\beta)$, β increases in b and $\lim_{b \rightarrow \infty} \beta = \infty$.
- (2) Without loss of generality, I focus on b sufficiently large such that $\underline{\theta} + b > \bar{\theta}$. Since $N_{no}^* = N(b)$, it follows that $\underline{\theta} + b > \bar{\theta}$ implies $N_{no}^* = 1$. Recall that without intervention the agent chooses $E[\theta] + b$ in equilibrium, and notice that $\underline{\theta} + b > \bar{\theta}$ implies that $E[\theta] + b > \bar{\theta}$. Therefore,

$$U_P^{**} = E[U_P(\theta, \theta)] - E[L(E[\theta] + b - \theta)].$$

- (3) Since $N_{in}^* \leq N_{no}^*$ and $N_{no}^* = 1$, we have $N_{in}^* = 1$ as well. Recall that with intervention the agent chooses $x^* > \underline{\theta} + \beta$, and notice that $\underline{\theta} + b > \bar{\theta}$ and $\beta > b$ imply $x^* > \bar{\theta}$. Based on Lemma A.1, $x^* - \theta > 0$ implies that $\Delta^*(x^* - \theta) > 0$ and $x^* - \theta - \Delta^*(x^* - \theta) > 0$. Therefore,

$$V_P^{**} = E[U_P(\theta, \theta)] - E[L(x^* - \theta - \Delta^*(x^* - \theta))] - \delta E[C(\Delta^*(x^* - \theta))].$$

Recall that $r - \Delta^*(r)$ and $\Delta^*(r)$ are increasing function of r . Since $x^* > \underline{\theta} + \beta$, we have

$$V_P^{**} \leq E[U_P(\theta, \theta)] - E[L(\underline{\theta} + \beta - \theta - \Delta^*(\underline{\theta} + \beta - \theta)) + \delta C(\Delta^*(\underline{\theta} + \beta - \theta))].$$

Therefore, if

$$\begin{aligned} E[L(E[\theta] - \theta + b)] &< E[L(E[\theta] - \theta - (E[\theta] - \underline{\theta}) + \beta - \Delta^*(E[\theta] - \theta \\ &\quad - (E[\theta] - \underline{\theta}) + \beta))] + E[\delta C(\Delta^*(E[\theta] - \theta - (E[\theta] - \underline{\theta}) + \beta))], \end{aligned}$$

it follows that $V_P^{**} < U_P^{**}$. A sufficient condition for $V_P^{**} < U_P^{**}$ is

$$\begin{aligned} L(y + b) &< L(y - (E[\theta] - \underline{\theta}) + \beta - \Delta^*(y - (E[\theta] - \underline{\theta}) + \beta)) \\ &\quad + \delta C(\Delta^*(y - (E[\theta] - \underline{\theta}) + \beta)) \end{aligned}$$

for every y .

- (4) I prove that $\lim_{r \rightarrow \infty} \Delta^*(r) = \lim_{r \rightarrow \infty} r - \Delta^*(r) = \infty$. Recall from Lemma A.1 that $r > 0$ implies that $\Delta^*(r) \in (0, r)$ and $L'(r - \Delta^*(r)) = \delta C'(\Delta^*(r))$. The latter can be rewritten as $r = \Delta^*(r) + (L')^{-1}(\delta C'(\Delta^*(r)))$. Recall that $C'' > 0$ and $L'' > 0$. Since $\Delta + (L')^{-1}(\delta C'(\Delta))$ is a strictly increasing, continuous, and unbounded function of Δ on $[0, \infty)$, the identity $r = \Delta^*(r) + (L')^{-1}(\delta C'(\Delta^*(r)))$ implies that $\lim_{r \rightarrow \infty} \Delta^*(r) = \infty$. The proof of Lemma 1 shows that $\lim_{r \rightarrow \infty} C''(r) > 0$ implies that $\lim_{r \rightarrow \infty} [r - \Delta^*(r)] = \infty$.

(5) I prove that for every y and $z > 0$,

$$\lim_{b \rightarrow \infty} \frac{L(y + \beta - \Delta^*(y + \beta)) + \delta C(\Delta^*(y + \beta))}{L(y - z + \beta - \Delta^*(y - z + \beta)) + \delta C(\Delta^*(y - z + \beta))} = 1.$$

Since $\lim_{b \rightarrow \infty} \beta = \infty$, step (4) implies that

$$\begin{aligned} \lim_{b \rightarrow \infty} \Delta^*(y - z + \beta) &= \lim_{b \rightarrow \infty} \Delta^*(y + \beta) = \infty \quad \text{and} \\ \lim_{b \rightarrow \infty} y - z + \beta - \Delta^*(y - z + \beta) &= \lim_{b \rightarrow \infty} y + \beta - \Delta^*(y + \beta) = \infty. \end{aligned}$$

Since $L', L'', C', C'' > 0$,

$$\begin{aligned} \lim_{b \rightarrow \infty} L(y + \beta - \Delta^*(y + \beta)) + \delta C(\Delta^*(y + \beta)) &= \\ \lim_{b \rightarrow \infty} L(y - z + \beta - \Delta^*(y - z + \beta)) + \delta C(\Delta^*(y - z + \beta)) &= \infty. \end{aligned}$$

From L'Hospital's Rule,

$$\begin{aligned} &\lim_{b \rightarrow \infty} \frac{L(y + \beta - \Delta^*(y + \beta)) + \delta C(\Delta^*(y + \beta))}{L(y - z + \beta - \Delta^*(y - z + \beta)) + \delta C(\Delta^*(y - z + \beta))} \\ &= \lim_{b \rightarrow \infty} \frac{L'(y + \beta - \Delta^*(y + \beta))(1 - \frac{\partial \Delta^*(y + \beta)}{\partial r} \frac{\partial \beta}{\partial b}) + \delta C'(\Delta^*(y + \beta)) \frac{\partial \Delta^*(y + \beta)}{\partial r} \frac{\partial \beta}{\partial b}}{L'(y - z + \beta - \Delta^*(y - z + \beta))(1 - \frac{\partial \Delta^*(y - z + \beta)}{\partial r} \frac{\partial \beta}{\partial b}) + \delta C'(\Delta^*(y - z + \beta)) \frac{\partial \Delta^*(y - z + \beta)}{\partial r} \frac{\partial \beta}{\partial b}} \\ &= \lim_{b \rightarrow \infty} \frac{\delta C'(\Delta^*(y + \beta))}{\delta C'(\Delta^*(y - z + \beta))}, \end{aligned}$$

where in the second equality, I used the identity $L'(r - \Delta^*(r)) = \delta C'(\Delta^*(r))$ from Lemma A.1. Similar to step (4) in the proof of part (i) with respect to δ , it follows that $\lim_{b \rightarrow \infty} \frac{\delta C'(\Delta^*(y + \beta))}{\delta C'(\Delta^*(y - z + \beta))} = 1$, as required.

- (6) I argue that $\lim_{b \rightarrow \infty} \frac{\partial \beta}{\partial b} > 1$. Note that $\beta = b + \Delta^*(\beta)$ implies that $\frac{\partial \beta}{\partial b} = 1 + \frac{\partial \Delta^*(\beta)}{\partial \beta} \frac{\partial \beta}{\partial b} \Rightarrow \frac{\partial \beta}{\partial b} = \frac{1}{1 - \frac{\partial \Delta^*(\beta)}{\partial \beta}}$. Therefore, $\lim_{b \rightarrow \infty} \frac{\partial \beta}{\partial b} = \lim_{b \rightarrow \infty} \frac{1}{1 - \frac{\partial \Delta^*(\beta)}{\partial \beta}}$. I argue that $\lim_{r \rightarrow \infty} \frac{\partial \Delta^*(r)}{\partial r} = d$ for some $d \in (0, 1)$. If true, then $\lim_{b \rightarrow \infty} \frac{\partial \beta}{\partial b} = \frac{1}{1-d} \in (1, \infty)$. Recall from step (4) that $\lim_{r \rightarrow \infty} \Delta^*(r) = \infty$ and $\lim_{r \rightarrow \infty} r - \Delta^*(r) = \infty$. Also recall from Lemma A.1 that $\frac{\partial \Delta^*(r)}{\partial r} \in (0, 1)$ for any finite r . If on the contrary $\lim_{r \rightarrow \infty} \frac{\partial \Delta^*(r)}{\partial r} = 0$, then it must be the case that $\lim_{r \rightarrow \infty} \Delta^*(r) < \infty$, a contradiction. If on the contrary $\lim_{r \rightarrow \infty} \frac{\partial \Delta^*(r)}{\partial r} = 1$, then it must be the case that $\lim_{r \rightarrow \infty} r - \Delta^*(r) < \infty$, a contradiction. Therefore, $\lim_{r \rightarrow \infty} \frac{\partial \Delta^*(r)}{\partial r} \in (0, 1)$, as required.
- (7) Based on the previous steps, $V_P^{**} < U_P^{**}$ holds if

$$\lim_{b \rightarrow \infty} \frac{L(y + b)}{L(y + \beta - \Delta^*(y + \beta)) + \delta C(\Delta^*(y + \beta))} < 1.$$

Since both the numerator and the denominator go to infinity as $b \rightarrow \infty$, L'Hospital's Rule implies that

$$\begin{aligned} & \lim_{b \rightarrow \infty} \frac{L(y+b)}{L(y+\beta-\Delta^*(y+\beta))+\delta C(\Delta^*(y+\beta))} \\ &= \lim_{b \rightarrow \infty} \frac{L'(y+b)}{L'(y+\beta-\Delta^*(y+\beta))(1-\frac{\partial \Delta^*(y+\beta)}{\partial r})\frac{\partial \beta}{\partial b}+\delta C'(\Delta^*(y+\beta))\frac{\partial \Delta^*(y+\beta)}{\partial r}\frac{\partial \beta}{\partial b}} \\ &= \lim_{b \rightarrow \infty} \frac{L'(y+b)}{L'(y+\beta-\Delta^*(y+\beta))\frac{\partial \beta}{\partial b}}. \end{aligned}$$

Since $\lim_{b \rightarrow \infty} \frac{\partial \beta}{\partial b} > 1$, it is sufficient to prove $\lim_{b \rightarrow \infty} \frac{L'(y+b)}{L'(y+b+\Delta^*(\beta)-\Delta^*(y+\beta))} \leq 1$. Notice that $L'' > 0$ and $\Delta^*(y+\beta) - \Delta^*(\beta) \leq y$ imply $\frac{L'(y+b)}{L'(y+b+\Delta^*(\beta)-\Delta^*(y+\beta))} \leq \frac{L'(y+b)}{L'(b)}$. Therefore, it is sufficient to prove

$$\lim_{b \rightarrow \infty} \frac{L'(y+b)}{L'(b)} = 1.$$

Define $\Lambda(r) \equiv r - \Delta^*(r)$ and notice that $\Lambda(r)$ is a strictly increasing and continuous function that spans $[0, \infty]$. Let r be such that $b = \Lambda(r)$. Then the identity $L'(r - \Delta^*(r)) = \delta C'(\Delta^*(r))$ in Lemma A.1 implies that

$$\begin{aligned} L'(b) &= L'(\Lambda(r)) \\ &= L'(r - \Delta^*(r)) \\ &= \delta C'(\Delta^*(r)) \\ &= \delta C'(r - \Lambda(r)) \\ &= \delta C'(\Lambda^{-1}(b) - b). \end{aligned}$$

Similarly, $L'(y+b) = \delta C'(\Lambda^{-1}(y+b) - (y+b))$. Therefore, $\frac{L'(y+b)}{L'(b)} = \frac{C'(\Lambda^{-1}(y+b) - (y+b))}{C'(\Lambda^{-1}(b) - b)}$. L'Hospital's Rule implies that

$$\begin{aligned} \lim_{b \rightarrow \infty} \frac{L'(y+b)}{L'(b)} &= \lim_{b \rightarrow \infty} \frac{C'(\Lambda^{-1}(y+b) - (y+b))}{C'(\Lambda^{-1}(b) - b)} \\ &= \lim_{b \rightarrow \infty} \frac{C''(\Lambda^{-1}(y+b) - (y+b)) \frac{1}{\Lambda'(\Lambda^{-1}(y+b))} - 1}{C''(\Lambda^{-1}(b) - b) \frac{1}{\Lambda'(\Lambda^{-1}(b))} - 1}. \end{aligned}$$

Notice that $\lim_{b \rightarrow \infty} \Lambda^{-1}(b) = \infty$ and $\Lambda' = 1 - \frac{\partial \Delta^*(r)}{\partial r}$. Since step (6) proves $\lim_{r \rightarrow \infty} \frac{\partial \Delta^*(r)}{\partial r} = d \in (0, 1)$, it must be the case that

$$\lim_{b \rightarrow \infty} \frac{\frac{1}{\Lambda'(\Lambda^{-1}(y+b))} - 1}{\frac{1}{\Lambda'(\Lambda^{-1}(b))} - 1} = \lim_{b \rightarrow \infty} \frac{\frac{1}{1-d} - 1}{\frac{1}{1-d} - 1} = 1.$$

If so, then

$$\lim_{b \rightarrow \infty} \frac{C'(\Lambda^{-1}(y+b) - (y+b))}{C'(\Lambda^{-1}(b) - b)} = \lim_{b \rightarrow \infty} \frac{C''(\Lambda^{-1}(y+b) - (y+b))}{C''(\Lambda^{-1}(b) - b)}.$$

Since there is a finite $\kappa \geq 2$ such that $\lim_{\Delta \rightarrow \infty} \frac{\partial^\kappa C(\Delta)}{\partial^\kappa \Delta} \in (-\infty, +\infty)$, by repeating this analysis κ times it must be the case that

$$\lim_{b \rightarrow \infty} \frac{C'(\Lambda^{-1}(y+b) - (y+b))}{C'(\Lambda^{-1}(b) - b)} = 1,$$

as required.

Consider part (ii). According to the proof of Theorem 1, if the partition has one element, then the agent chooses $E[\theta] + b$ without intervention, otherwise $x^* \in (\underline{\theta} + \beta, \bar{\theta} + \beta)$. Therefore,

$$U_P^* = E[U_P(\theta, \theta)] - E[L(|E[\theta] + b - \theta|)]$$

and

$$\begin{aligned} V_P^* &= E[U_P(\theta, \theta)] - E[\min_{\Delta} \{L(|x^* - \theta - \Delta|) + \delta C(|\Delta|)\}] \\ &\geq E[U_P(\theta, \theta)] - E[\min_{\Delta} \{L(|\bar{\theta} + \beta - \theta - \Delta|) + \delta C(|\Delta|)\}]. \end{aligned}$$

Suppose that $b < \frac{\bar{\theta} - \underline{\theta}}{4}$. Based on equation (12), $b < \frac{\bar{\theta} - \underline{\theta}}{4}$ implies that $N(b) \geq 2$, that is, $N_{no}^* \geq 2$. Define $\underline{\delta}(b)$ such that $\beta(\underline{\delta}(b)) = \bar{\theta} - \underline{\theta}$. Since $b < \frac{\bar{\theta} - \underline{\theta}}{4}$, $\frac{\partial \beta}{\partial \delta} < 0$, $\lim_{\delta \rightarrow \infty} \beta = b$, and $\lim_{\delta \rightarrow 0} \beta = \infty$, $\underline{\delta}(b)$ is uniquely defined. Also, since $\frac{\partial \beta}{\partial b} > 0$, $\underline{\delta}(\cdot)$ is an increasing and continuous function. Finally, since $\lim_{b \rightarrow 0} \beta = 0$ and $\lim_{\delta \rightarrow 0} \beta = \infty$, it must be that $\lim_{b \rightarrow 0} \underline{\delta}(b) = 0$. Note that $\beta(\underline{\delta}(b)) = \bar{\theta} - \underline{\theta}$ implies that $N_{in}^* = 1$. Suppose that $\delta = \underline{\delta}(b)$ and let $\Delta(\rho, \theta) \equiv \frac{3}{2}(\bar{\theta} - \underline{\theta}) + \rho(E[\theta] - \theta)$ for some $\rho \in (0, 1)$. Then

$$\begin{aligned} V_P^* &\geq E[U_P(\theta, \theta)] - E[L(|\bar{\theta} + \beta(\underline{\delta}(b)) - \theta - \Delta(\rho, \theta)|) + \underline{\delta}(b)C(|\Delta(\rho, \theta)|)] \\ &= E[U_P(\theta, \theta)] - E[L(|\bar{\theta} + \bar{\theta} - \underline{\theta} - \theta - \frac{3}{2}(\bar{\theta} - \underline{\theta}) - \rho(E[\theta] - \theta)|) + \underline{\delta}(b)C(|\Delta(\rho, \theta)|)] \\ &= E[U_P(\theta, \theta)] - E[L(|(1 - \rho)(E[\theta] - \theta)|) + \underline{\delta}(b)C(|\Delta(\rho, \theta)|)]. \end{aligned}$$

Since $\lim_{b \rightarrow 0} \underline{\delta}(b) = 0$, we have

$$\begin{aligned} \lim_{b \rightarrow 0} V_P^* &\geq E[U_P(\theta, \theta)] - E[L(|(1 - \rho)(E[\theta] - \theta)|)] \\ &> E[U_P(\theta, \theta)] - E[L(|E[\theta] - \theta|)] \\ &= \lim_{b \rightarrow 0} U_P^*. \end{aligned}$$

Therefore, there exists $\underline{b} \in (0, \frac{\bar{\theta} - \underline{\theta}}{4})$ such that if $b \in (0, \underline{b})$ and $\delta = \underline{\delta}(b)$, then $U_P^* < V_P^*$. Since $N_{no}^* \geq 2$, from Crawford and Sobel (1982) we know that $U_P^* < U_P^{**}$. Recall that β is decreasing in δ , and for a given b , $\lim_{\delta \rightarrow 0} V_P^* = -\infty$. Therefore,

without loss of generality we can choose $\underline{\delta}(b)$ sufficiently small such that $N_{in}^* = 1$, $N_{no}^* \geq 2$, and $U_P^* < V_P^* < U_P^{**}$. Based on part (i), it must be the case that $\underline{\delta}(b) < \bar{\delta}$. Finally, from continuity, there is an interval of values of δ that satisfy these conditions. ■

PROOF OF PROPOSITION 2: I first prove the result with respect to δ . Recall that $\lim_{\delta \rightarrow 0} \beta = \infty$. Therefore, without loss of generality suppose δ is sufficiently small such that $N_{in}^* = 1$. Then

$$\begin{aligned} V_A^{**} &= E[U_A(\theta + b, \theta; b)] - \min_x E[T(|x - \theta - \Delta^*(x - \theta) - b|)] \\ &\geq E[U_A(\theta + b, \theta; b)] - E[T(|E[\theta] + \beta - \theta - \Delta^*(E[\theta] + \beta - \theta) - b|)] \\ &= E[U_A(\theta + b, \theta; b)] - E[T(|E[\theta] - \theta + \Delta^*(\beta) - \Delta^*(E[\theta] - \theta + \beta)|)]. \end{aligned}$$

The inequality follows from the observation that the agent can always do weakly better than choosing $x = E[\theta] + \beta$, and the second equality follows from $\beta = b + \Delta^*(\beta)$. Note that based on the proof of Lemma A.1,

$$\frac{\partial \Delta^*}{\partial r} = \frac{L''(|r - \Delta^*|)}{L''(|r - \Delta^*|) + \delta C''(|\Delta^*|)}.$$

Also notice that $\lim_{\delta \rightarrow 0} \Delta^*(r) = r$, and because $L''(0) > 0$,

$$\lim_{\delta \rightarrow 0} \frac{\partial \Delta^*}{\partial r} = \frac{L''(|r - r|)}{L''(|r - r|) + \lim_{\delta \rightarrow 0} \delta C''(|r|)} = \frac{L''(0)}{L''(0)} = 1.$$

Since $\lim_{\delta \rightarrow 0} \frac{\partial \Delta^*}{\partial r} = 1$, it must be the case that

$$\lim_{\delta \rightarrow 0} [\Delta^*(\beta) - \Delta^*(E[\theta] - \theta + \beta)] = \theta - E[\theta],$$

which implies that

$$\begin{aligned} \lim_{\delta \rightarrow 0} E[T(|E[\theta] - \theta + \Delta^*(\beta) - \Delta^*(E[\theta] - \theta + \beta)|)] &= \lim_{\delta \rightarrow 0} E[T(|E[\theta] - \theta + \theta - E[\theta]|)] \\ &= T(0) = 0. \end{aligned}$$

Therefore, $\lim_{\delta \rightarrow 0} V_A^{**} = E[U_A(\theta + b, \theta; b)]$, the highest expected utility that the agent can get. Notice that as long as $b > 0$, in the game without intervention communication is always imperfect, and therefore, $U_A^{**} < E[U_A(\theta + b, \theta; b)]$. Moreover, note that U_A^{**} is independent of δ . Therefore, there exists $\bar{\delta} > 0$, as required.

Next, I prove the result with respect to b . Recall that β increases in b and $\lim_{b \rightarrow \infty} \beta = \infty$. Similar to the proof of Proposition 1, if b is sufficiently large, then $N_{no}^* = N_{in}^* = 1$. As above,

$$V_A^{**} \geq E[U_A(\theta + b, \theta; b)] - E[T(|E[\theta] - \theta + \Delta^*(\beta) - \Delta^*(E[\theta] - \theta + \beta)|)].$$

Recall that if $N_{no}^* = 1$, then the agent chooses $E[\theta] + b$ in equilibrium without intervention, which implies that

$$\begin{aligned} U_A^{**} &= E[U_A(\theta + b, \theta; b)] - E[T(|E[\theta] + b - (\theta + b)|)] \\ &= E[U_A(\theta + b, \theta; b)] - E[T(|E[\theta] - \theta|)]. \end{aligned}$$

Therefore, if

$$E[T(|E[\theta] - \theta + \Delta^*(\beta) - \Delta^*(E[\theta] + \beta - \theta)|)] < E[T(|E[\theta] - \theta|)], \quad (\text{A.13})$$

then $V_A^{**} > U_A^{**}$. Define $\Gamma(x) \equiv x + \Delta^*(\beta) - \Delta^*(x + \beta)$. Notice that $\Gamma(0) = 0$ and $\Gamma'(x) = 1 - \frac{\partial \Delta^*(x + \beta)}{\partial x} \in (0, 1)$. Therefore, if $x > 0$, then $\Gamma(x) < x$, and if $x < 0$, then $\Gamma(x) > x$. In other words, $|\Gamma(x)| < |x|$. If so, $T(|\Gamma(x)|) < T(|x|)$, which implies that $T(|\Gamma(E[\theta] - \theta)|) < T(|E[\theta] - \theta|)$ for a given θ . Therefore, $E[T(|\Gamma(E[\theta] - \theta)|)] < E[T(|E[\theta] - \theta|)]$, which implies inequality (A.13), as required. ■

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication code.